Intermediate Statistics using SPSS

EPSY 5383

Class Notes

Correlation

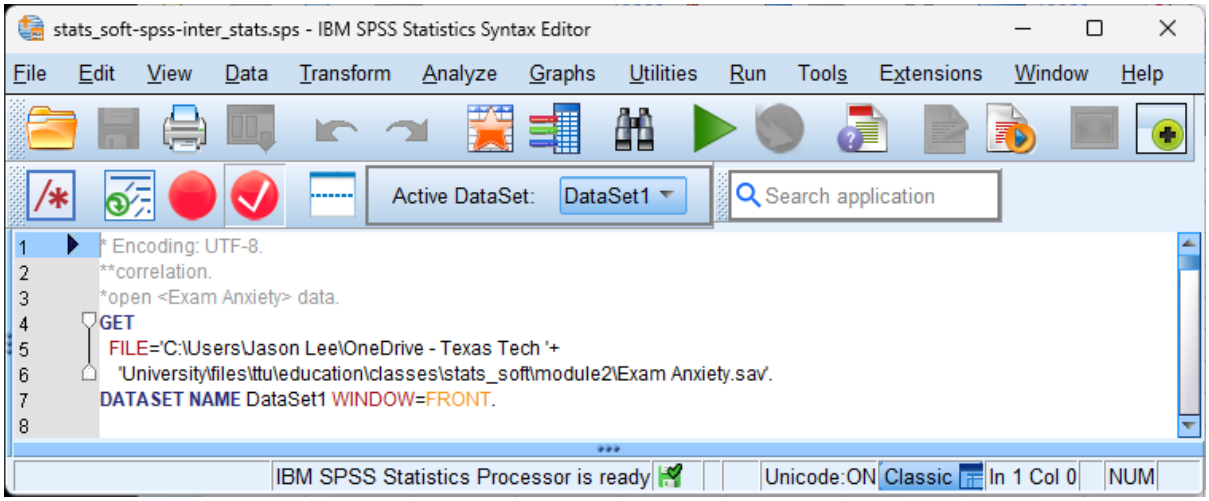
Goal: Compute correlation and interpret the results.

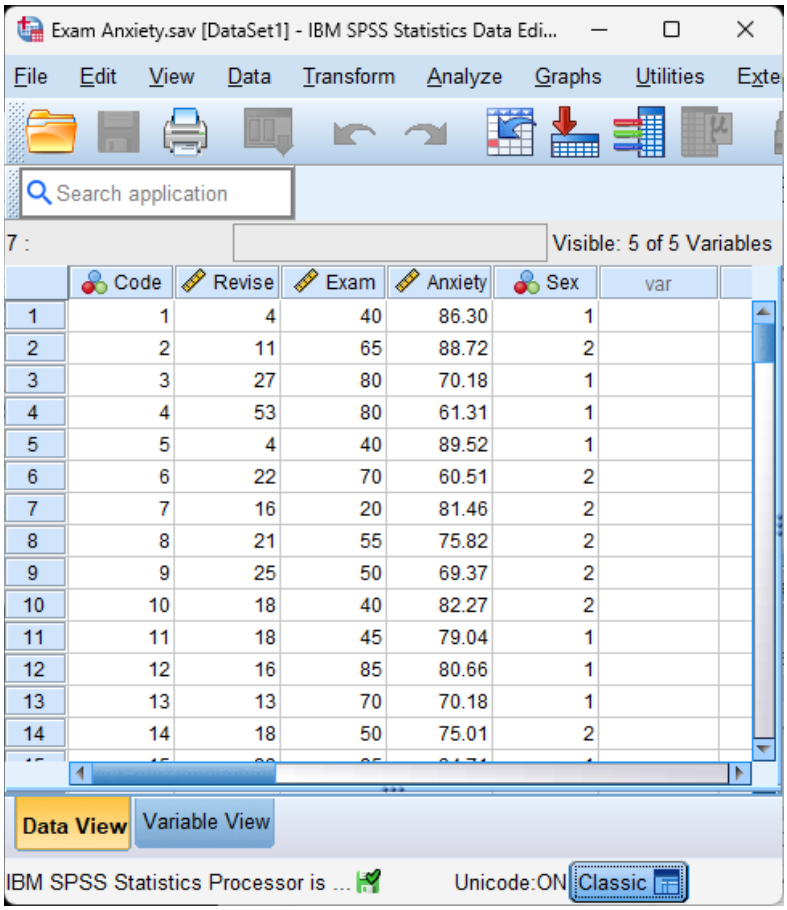
OVERVIEW

1. Watch the video <stats-correlation.mp4> (10:49)
2. Three well-known correlation coefficients:
    1. **Pearson's product-moment** correlation coefficient measures the linear relationship between two variables.
    2. **Spearman's rank** correlation coefficient, which applies Pearson's correlation to data converted into *ranked scores*, proves useful especially when the data does not exhibit a linear pattern or when working with ordinal or non-parametric data.
    3. **Kendall's tau** correlation coefficient, while similar to Spearman's correlation, is often favored when dealing with small data sets that feature a *large number of tied ranks*.

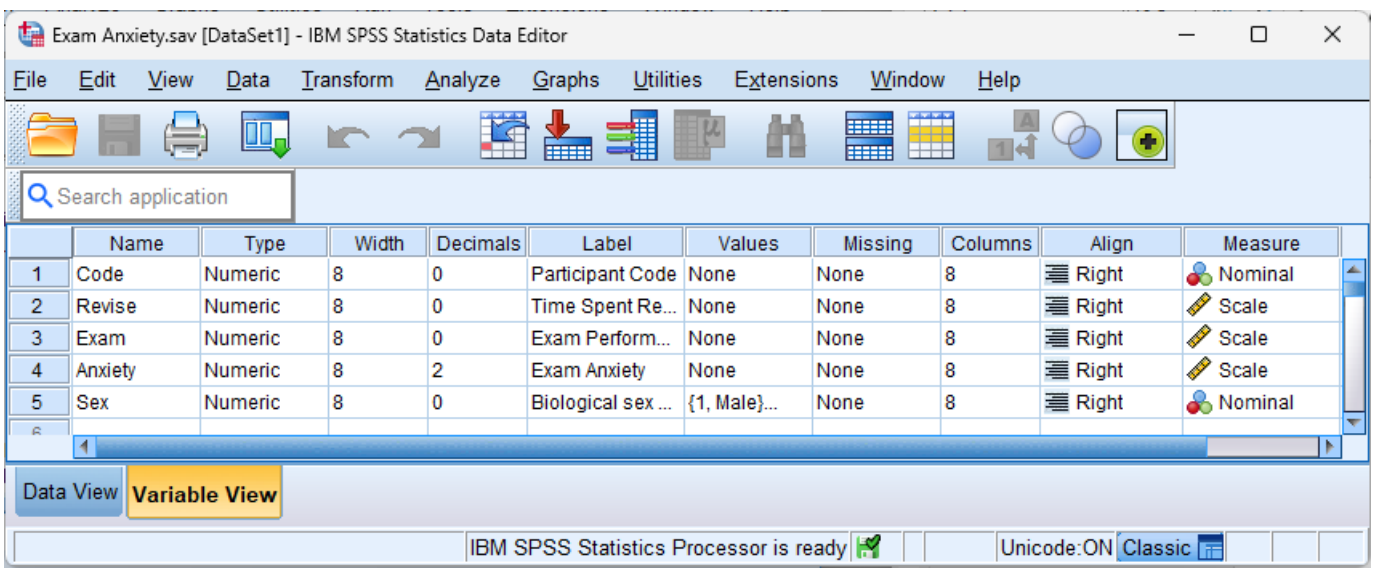
CREATE SYNTAX FILE & OPEN DATA FILE

3. From the top menu of the Data Editor, **File** >> **Open** >> **Data** >> Go to the directory containing the "Exam Anxiety.sav" data file and select the file >> **Paste**. Upon pasting, the syntax will auto-populate into the Syntax Editor.
4. From the top menu of the Syntax Editor, **File** >> **Save as** >> Type "stats_soft-spss-inter_stats" into the File name box >> **Save**.
5. To execute the generated syntax, select the syntax by 'dragging' your mouse over it and click the green triangle button on the top menu bar. The data will then appear in the Data Editor.





	Code	Revise	Exam	Anxiety	Sex	var
1	1	4	40	86.30	1	
2	2	11	65	88.72	2	
3	3	27	80	70.18	1	
4	4	53	80	61.31	1	
5	5	4	40	89.52	1	
6	6	22	70	60.51	2	
7	7	16	20	81.46	2	
8	8	21	55	75.82	2	
9	9	25	50	69.37	2	
10	10	18	40	82.27	2	
11	11	18	45	79.04	1	
12	12	16	85	80.66	1	
13	13	13	70	70.18	1	
14	14	18	50	75.01	2	



	Name	Type	Width	Decimals	Label	Values	Missing	Columns	Align	Measure
1	Code	Numeric	8	0	Participant Code	None	None	8	Right	Nominal
2	Revise	Numeric	8	0	Time Spent Re...	None	None	8	Right	Scale
3	Exam	Numeric	8	0	Exam Perform...	None	None	8	Right	Scale
4	Anxiety	Numeric	8	2	Exam Anxiety	None	None	8	Right	Scale
5	Sex	Numeric	8	0	Biological sex ...	{1, Male}...	None	8	Right	Nominal

a. A psychologist is exploring the impact of exam stress and study time on test results.

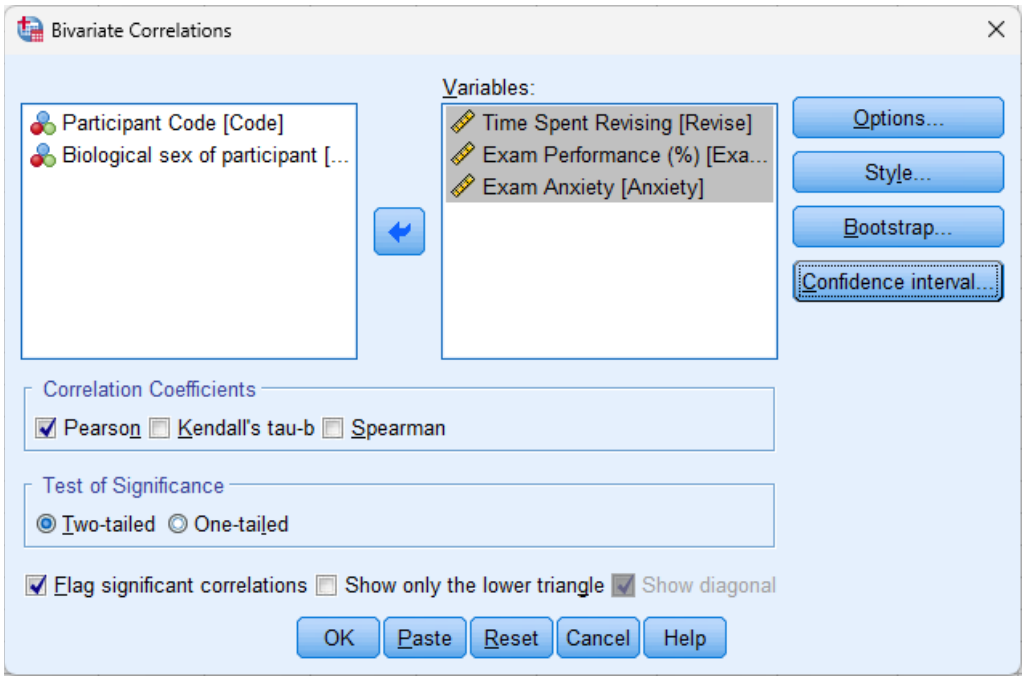
b. She developed an instrument to gauge exam-related stress, named the Exam Anxiety Questionnaire.

c. The level of anxiety was assessed prior to the students' exams, and their performance was evaluated based on their percentage score.

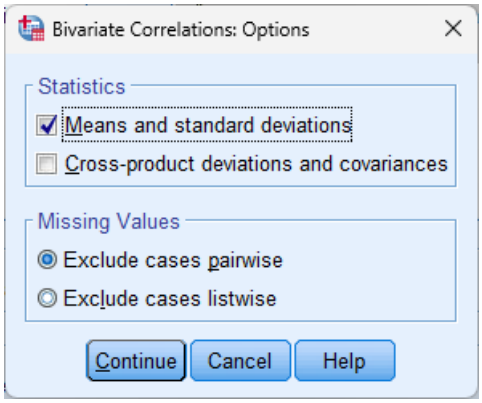
d. Additionally, the amount of time students dedicated to studying was also tracked.

CALCULATE PEARSON CORRELATION

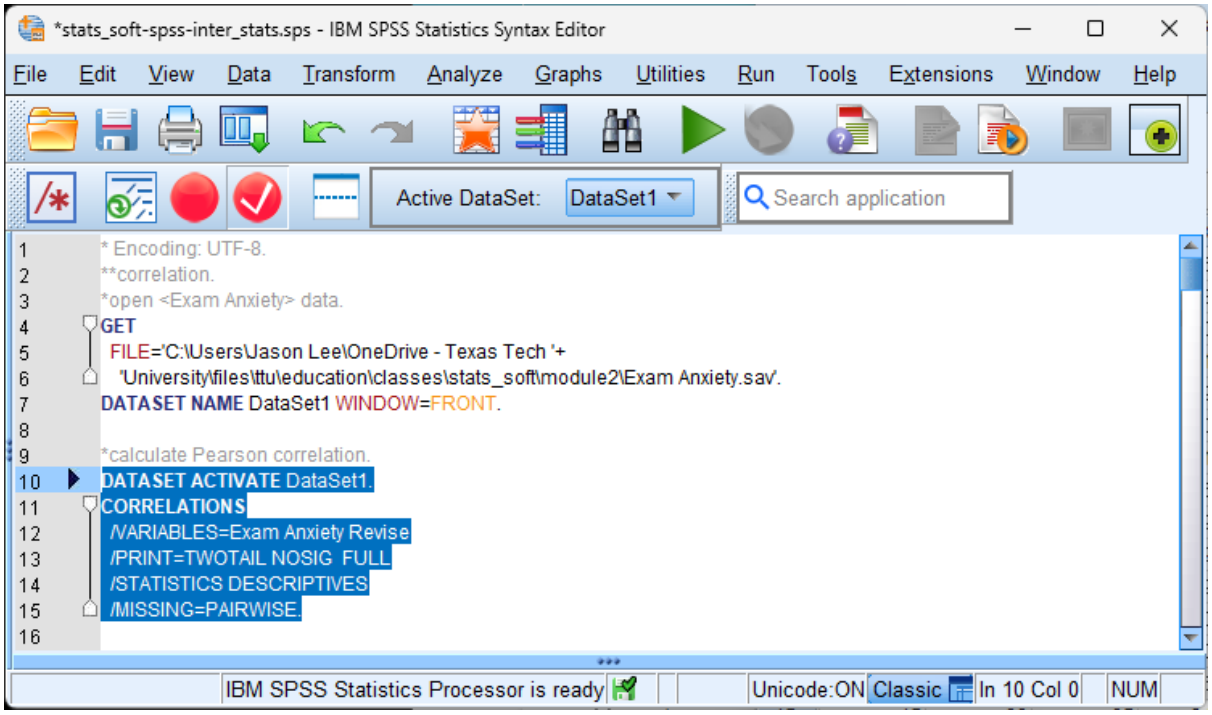
6. From the top menu (of either the Data Editor or Syntax Editor), **Analyze** >> **Correlate** >> **Bivariate**. This will open the Bivariate Correlations window.
7. In the left box of the window, select the three variables Exam performance, Exam anxiety, and Time Spent Revising >> Move these variables into the right Variables box by either (a) dragging, (b) 'double' clicking, or (c) clicking the arrow button .
8. At the bottom section of the window, select **Pearson**, **Two-tailed**, and **Flag significant correlations** options.



9. **Options** >> Select **Mean and standard deviations** option >> **Continue**.



10. **Paste** >> Go to the Syntax Editor, **Save**.
11. Execute the generated syntax.



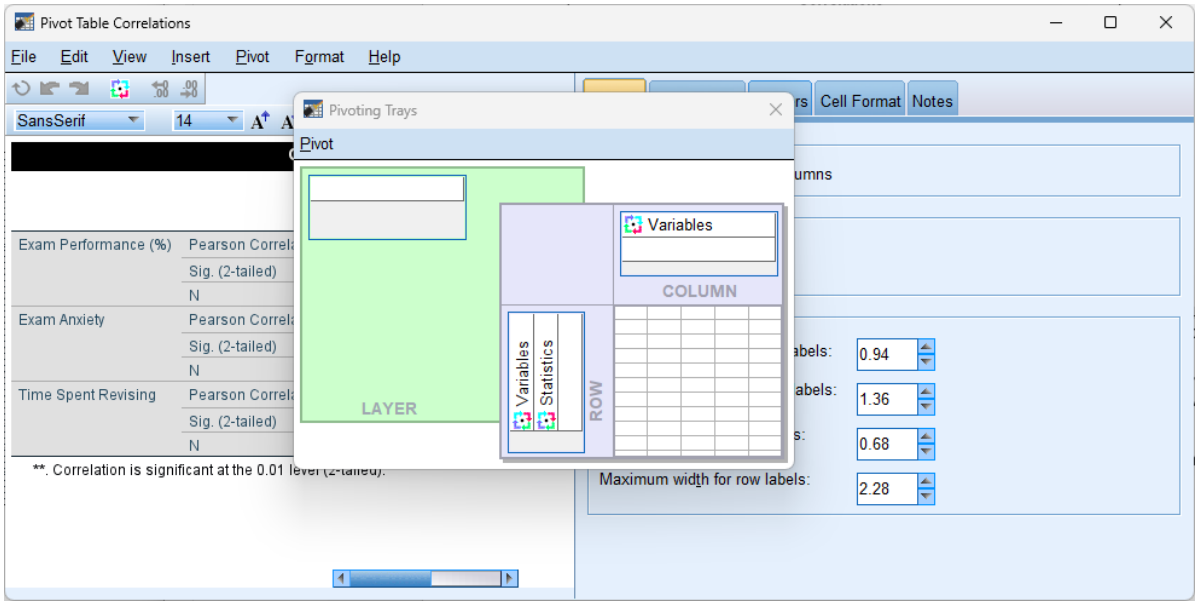
INTERPRET RESULTS

Correlations

		Exam Performance (%)	Exam Anxiety	Time Spent Revising
Exam Performance (%)	Pearson Correlation	1	-.441**	.397**
	Sig. (2-tailed)		<.001	<.001
	N	103	103	103
Exam Anxiety	Pearson Correlation	-.441**	1	-.709**
	Sig. (2-tailed)	<.001		<.001
	N	103	103	103
Time Spent Revising	Pearson Correlation	.397**	-.709**	1
	Sig. (2-tailed)	<.001	<.001	
	N	103	103	103

**. Correlation is significant at the 0.01 level (2-tailed).

12. The correlation between exam anxiety and performance is significant ($r = -.44$, $p < .001$), indicating that higher stress levels are associated with lower exam scores.
13. The correlation between the time spent revising and exam performance is significant ($r = .40$, $p < .001$), indicating that students who invest more time in preparation tend to score higher on exams.
14. To modify the table format and content display, 'double' click on the **Correlations** table and then click the **Pivoting Trays** icon from the top menu bar in the Pivot Table Correlations window.



End of Demonstration

Simple linear regression

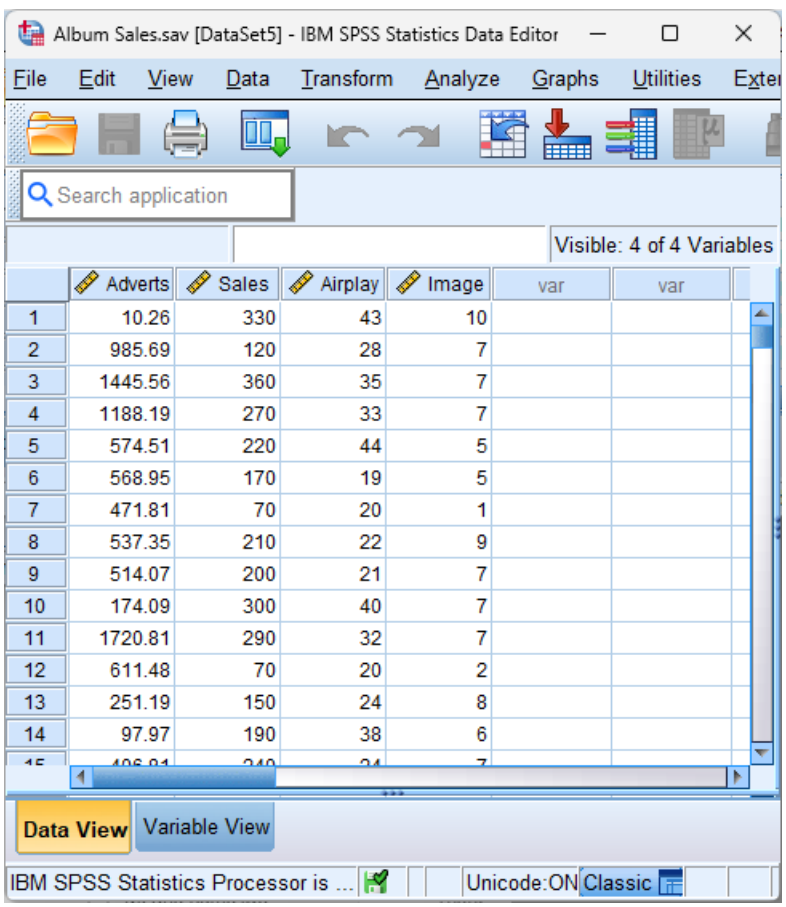
Goal: Run simple linear regression and interpret the results.

OVERVIEW

1. Watch the video <stats-simple_regression.mp4> (13:54)
2. Linear models make the following assumptions:
    a. *Linearity*: The predictor variable and the outcome variable have a linear relationship.
    b. *Normality*: To have optimal *b* estimates, the residuals should follow a normal distribution. For accurate CIs, the sampling distribution of *b* should also be normal.
    c. *Homoscedasticity*: The variance of the residuals should remain relatively constant as the value of the predictor changes. This is essential for optimal *b* estimates, and for accuracy in significance tests and CIs of the parameters.

OPEN SYNTAX FILE AND DATA FILE

3. Open the "stats_soft-spss-inter_stats.sps" syntax file.
4. **File** >> **Open** >> **Data** >> Go to the directory containing the "Album Sales.sav" data file and select the file >> **Paste** >> **Save** and execute the generated syntax.



	Adverts	Sales	Airplay	Image	var	var
1	10.26	330	43	10		
2	985.69	120	28	7		
3	1445.56	360	35	7		
4	1188.19	270	33	7		
5	574.51	220	44	5		
6	568.95	170	19	5		
7	471.81	70	20	1		
8	537.35	210	22	9		
9	514.07	200	21	7		
10	174.09	300	40	7		
11	1720.81	290	32	7		
12	611.48	70	20	2		
13	251.19	150	24	8		
14	97.97	190	38	6		

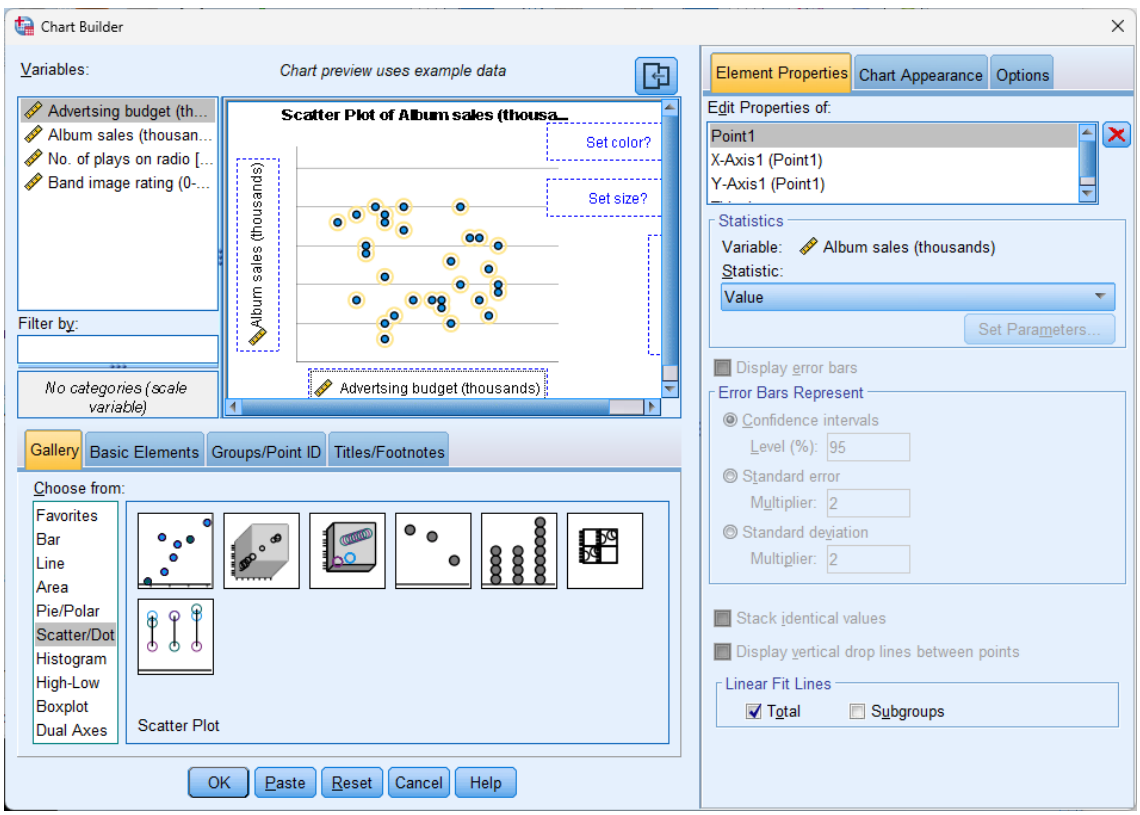
a. The Data View window displays 200 rows, with each row corresponding to a unique album.

b. Two columns (variables) are presented: one reflects the pre-release promotional expenditure for the album in pounds (Adverts), and the other represents the album sales in thousands after release (Sales).

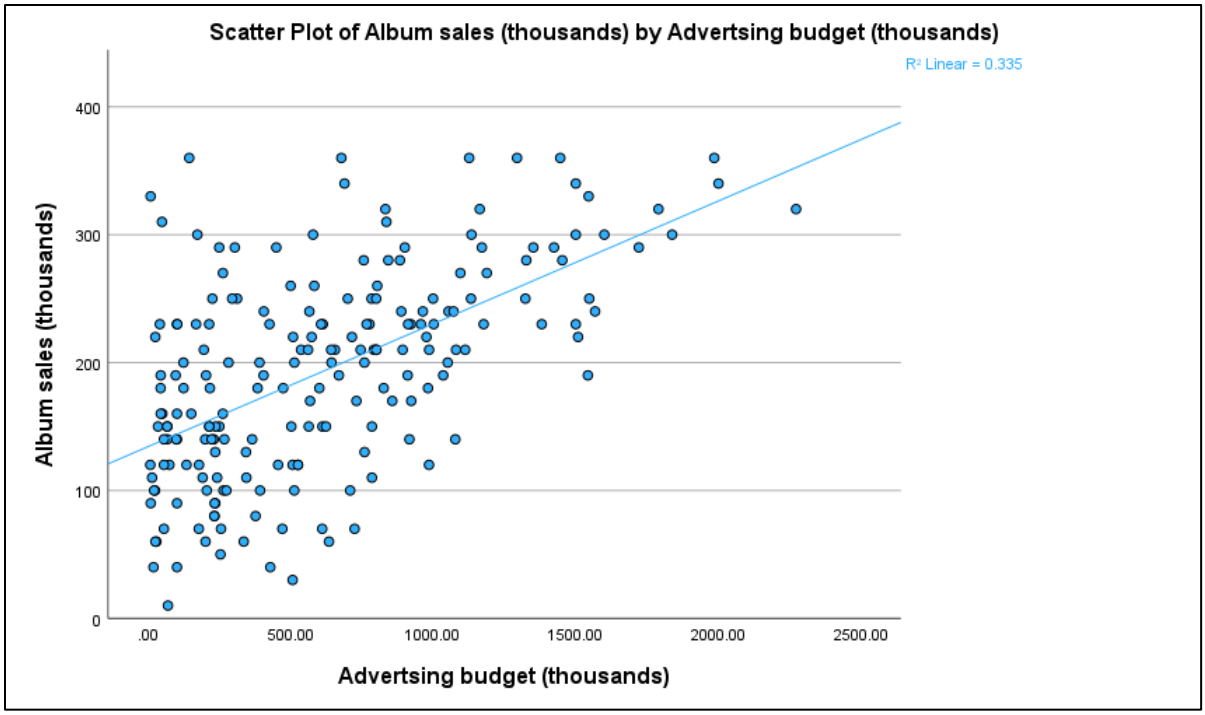
c. This layout is standard for entering regression data. The outcome variable and any predictor variables should be allocated to separate columns, while each row should contain independent values for those variables.

CREATE SCATTER PLOT

5. From the top menu, **Graph** >> **Chart Builder**. This will open the Chart Builder window.
6. In the lower section of the window, click **Gallery** tab >> From the Chose from menu, select **Scatter/Dot** and 'double' click on **Scatter plot**.
7. In the top section, drag the Sales variable from the Variables box to the Y-Axis? box on the graph.
8. Similarly, drag the Adverts variable from the Variables box into the X-Axis? box on the graph.
9. From the Linear Fit Lines menu on the right section of the window, choose **Total** option.



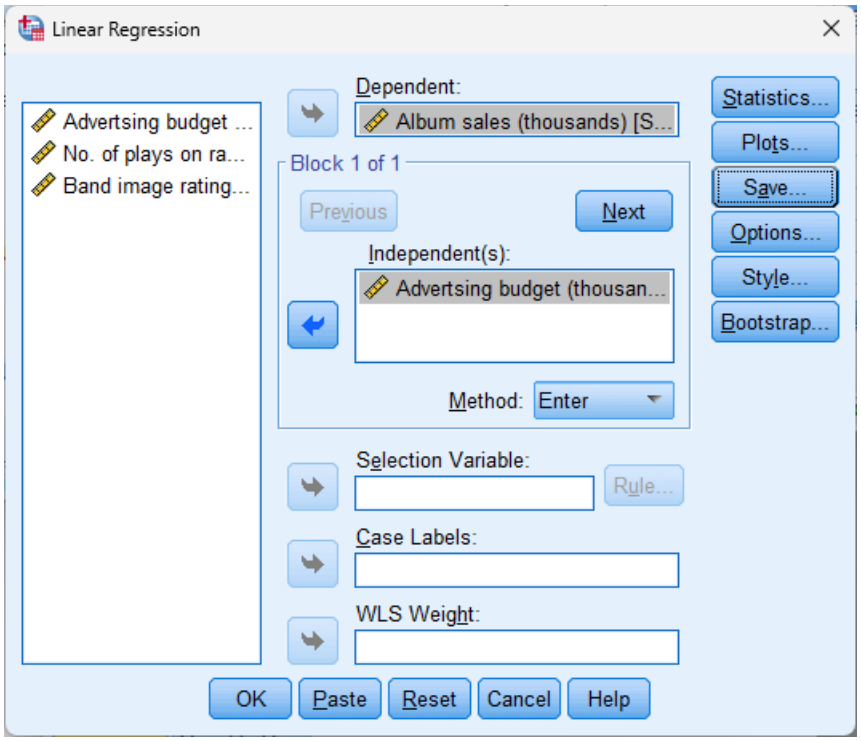
10. **Paste** >> **Save** and run the syntax.

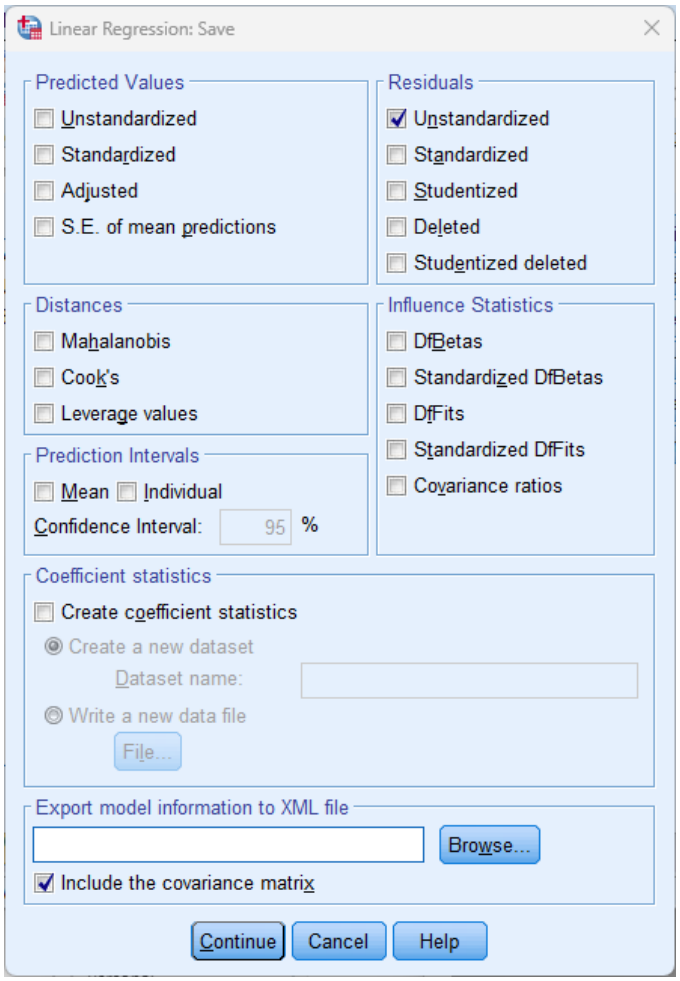


11. It is evident that a positive “linear” relationship exists. Thus, higher advertising expenditure for the album likely leads to increased sales. The scatter plot also includes the line of best fit for this data.
12. To determine the parameters that define the regression line and assess whether this line provides a useful model, a regression analysis must be conducted.

CONDUCT REGRESSION

13. **Analyze** >> **Regression** >> **Linear**. This will open the Linear Regression window.
14. In the left box of the window, select the variable Sales >> Move this variable into the Dependent box.
15. Select Adverts and move this variable into the Independent(s) box.
16. **Save** >> Select **Unstandardized** Residuals option >> **Continue**.
17. **Paste** >> **Save** and execute the generated syntax.


Linear Regression
Advertsing budget ...
No. of plays on ra...
Band image rating...
Dependent:
Album sales (thousands) [S...
Block 1 of 1
Previous
Next
Independent(s):
Advertsing budget (thousan...
Method: Enter
Selection Variable:
Rule...
Case Labels:
WLS Weight:
Statistics...
Plots...
Save...
Options...
Style...
Bootstrap...
OK
Paste
Reset
Cancel
Help


Linear Regression: Save
Predicted Values
Unstandardized
Standardized
Adjusted
S.E. of mean predictions
Residuals
Unstandardized
Standardized
Studentized
Deleted
Studentized deleted
Distances
Mahalanobis
Cook's
Leverage values
Influence Statistics
DfBetas
Standardized DfBetas
DfFits
Standardized DfFits
Covariance ratios
Prediction Intervals
Mean
Individual
Confidence Interval: 95 %
Coefficient statistics
Create coefficient statistics
Create a new dataset
Dataset name:
Write a new data file
File...
Export model information to XML file
Browse...
Include the covariance matrix
Continue
Cancel
Help

INTERPRET RESULTS

Model Summary[b]

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate
1	.578[a]	.335	.331	65.991

a. Predictors: (Constant), Advertsing budget (thousands)

b. Dependent Variable: Album sales (thousands)

18. The regression model accounts for 33.5% of the variance in album sales.

ANOVA[a]

Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	433687.833	1	433687.833	99.587	<.001[b]
	Residual	862264.167	198	4354.870		
	Total	1295952.000	199			

a. Dependent Variable: Album sales (thousands)

b. Predictors: (Constant), Advertsing budget (thousands)

19. The model significantly predicts album sales, $F(1, 198) = 99.59$, $p < .001$.

20. The **Model Summary** and **ANOVA** tables provide information about the *overall* predictive ability of the model for the outcome variable. However, they do not detail the individual contribution of variables within the model.

Coefficients[a]

Model		Unstandardized Coefficients B	Unstandardized Coefficients Std. Error	Standardized Coefficients Beta	t	Sig.
1	(Constant)	134.140	7.537		17.799	<.001
	Advertsing budget (thousands)	.096	.010	.578	9.979	<.001

a. Dependent Variable: Album sales (thousands)

21. The **Coefficients** table can be written into the following formula:

$$\text{Sales}_i = 134.14 + 0.096(\text{Adverts}_i)$$

22. The intercept *a* (Constant) indicates the value of the outcome variable when the value of the predictor variable is equal to 0. Thus, even with no advertising expenditure (Adverts = 0), we can expect that 134,140 albums will be sold.

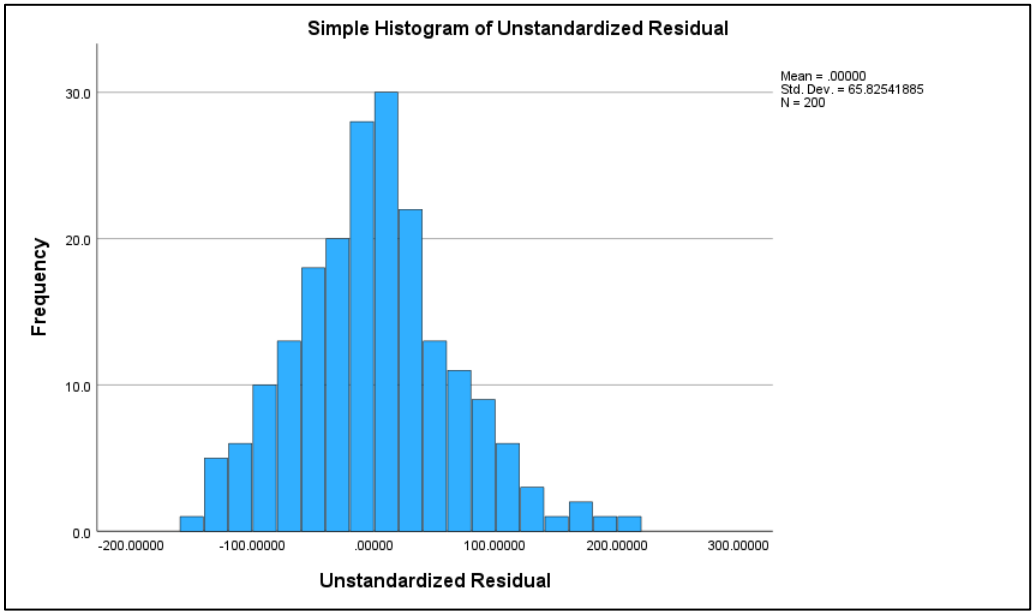
23. The advertising budget is a significant predictor of album sales ($p < .001$).

24. The regression coefficient *b* depicts the change in the outcome variable due to a unit change in the predictor variable. Therefore, for every £1,000 increase in advertising, we can expect that 96 (= 0.096 x 1,000) additional albums will be sold.

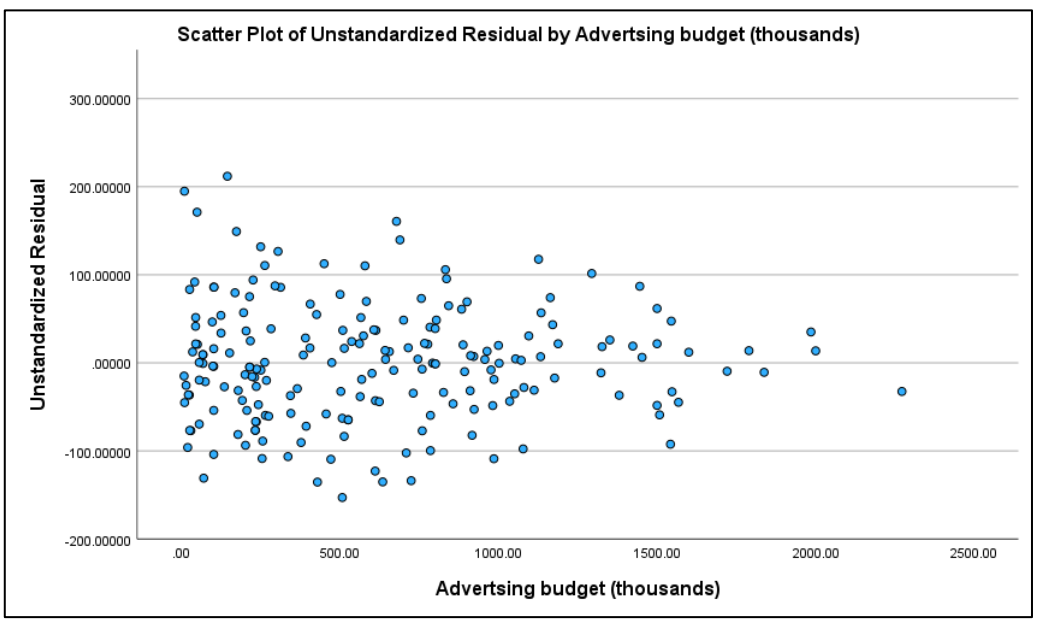
CHECK ASSUMPTIONS

25. *Linearity* has already been confirmed through the scatter plot.

26. By using the **Chart Builder** functions (>> **Histogram** >> **Simple Histogram**), make a histogram of the residuals that have been added to the data in a new column labeled RES_1. It is evident from the histogram that the residuals follow a normal distribution, therefore meeting the assumption of *normality*.



27. To assess *homoscedasticity*, generate a scatter plot with RES_1 on the y-axis and Adverts on the x-axis (refer to steps #5-11). The assumption of homoscedasticity appears unmet as the residuals display greater variability when the value of the predictor variable is smaller.

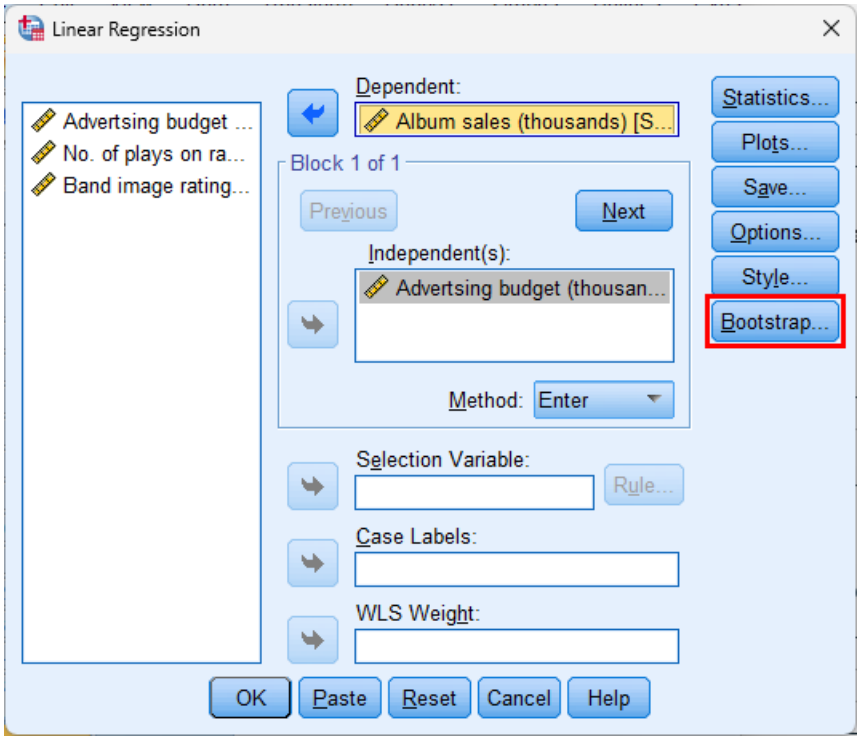


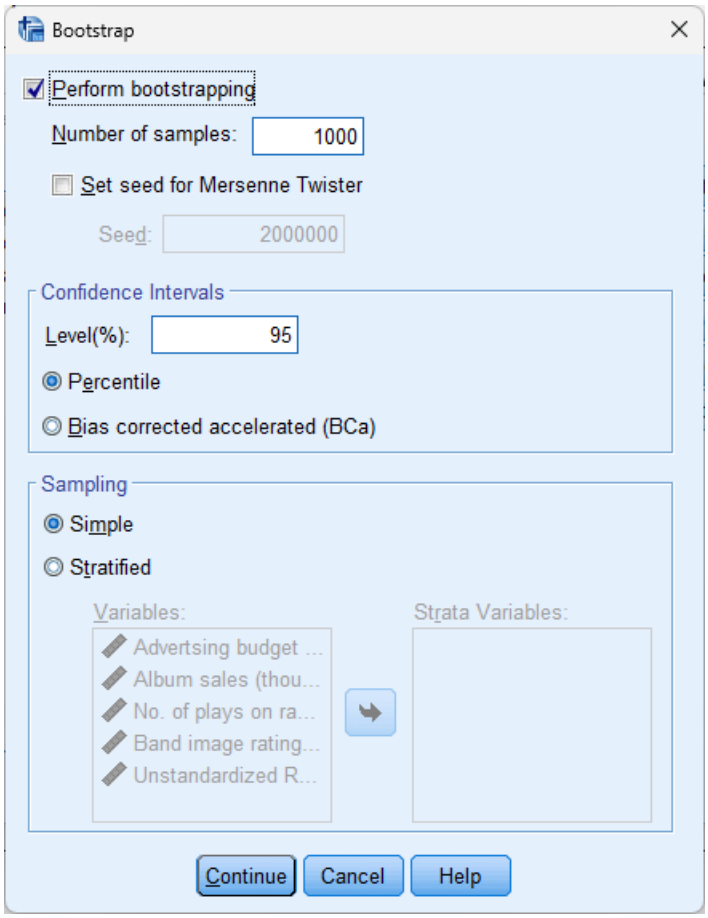
28. In light of the failure to meet the assumption of homoscedasticity, what are our next steps? Should we consider data transformations? What other remedies might be applicable?

End of Demonstration

Practice

1. If the assumptions of linearity, normality, and homoscedasticity are satisfied, then we can trust our *b* estimates. This allows us to generalize our model, implying it is applicable to other samples, not just the one from which we collected the data.
2. If these assumptions are potentially violated, we can employ **bootstrapping** (Efron & Tibshirani, 1993) to compute robust estimates for *b* and their confidence intervals (CI).
3. In the Linear Regression window, **Bootstrap** >> Select **Perform bootstrapping** >> Specify the **Number of samples** and the type of **Confidence intervals** and their level.





4. Watch the video <stats-bootstrapping.mp4> (9:43).

End of Practice

Multiple linear regression

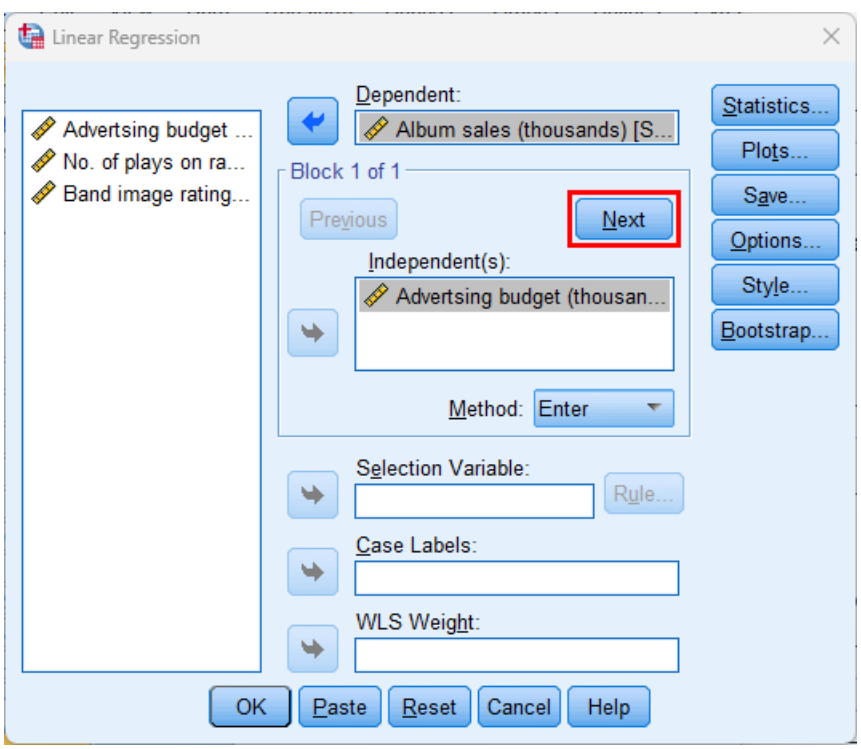
Goal: Run hierarchical, multiple linear regression and interpret the results.

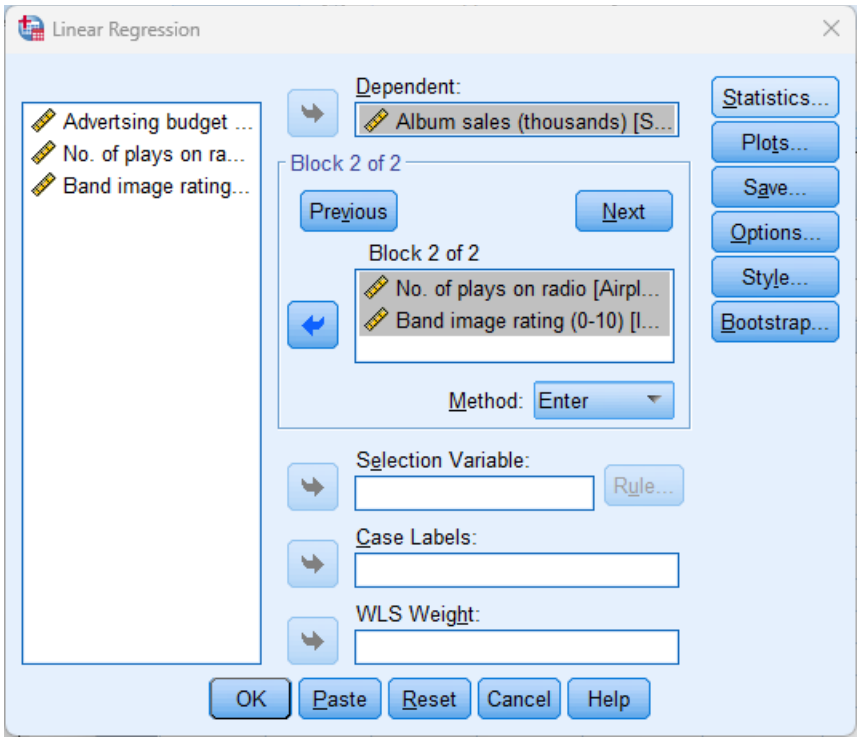
OVERVIEW

1. In the previous example, the executive determined that the advertising budget significantly predicts album sales, hence, this variable should be the first inclusion in the model.
2. Two additional variables should be taken into account: the number of radio plays (Airplay) and the band's attractiveness (Image).
3. These new variables would be introduced into the model *subsequent* to the advertising budget. This **hierarchical** method allows the researcher to determine the order of variable entry into the model, guided by previous research.
4. It is necessary to introduce variables in blocks, with each block representing a different step in the hierarchical sequence.

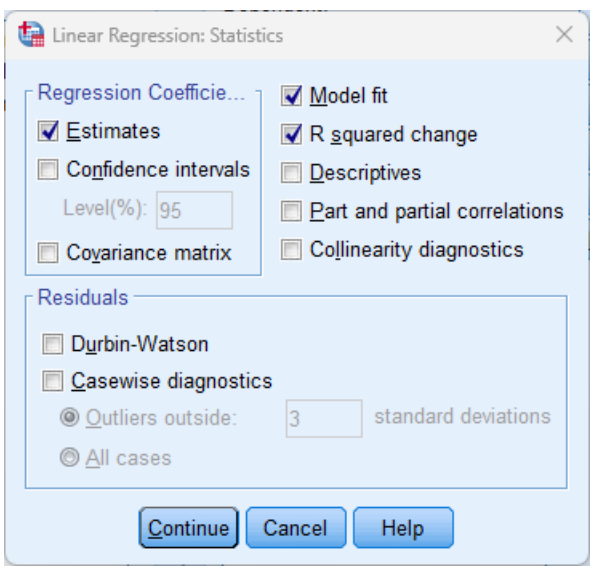
CONDUCT REGRESSION

5. **Analyze** >> **Regression** >> **Linear**. This will open the Linear Regression window.
6. To configure the first block, replicate the previous steps. In the left box of the window, select the variable Sales >> Move this variable into the Dependent box.
7. After defining the first block in the hierarchy, we will arrange the second block using the other two variables. Click **Next**. This action clears the Independent(s) box, allowing for the introduction of new predictor variables >> Select Airplay and Image and move these variables into the Independent(s) box.





8. Use the **Previous** and **Next** buttons to navigate between blocks and make modifications.
9. **Statistics** >> Add **R square change** option >> **Continue**.



10. **Paste** >> **Save** and execute the generated syntax.

INTERPRET RESULTS

11. In the **Model Summary** table, Model 1 pertains to the first block in the hierarchy when only advertising budget is used as the predictor variable. Model 2 pertains to when all three predictor variables are used.

Model Summary

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate	Change Statistics				
					R Square Change	F Change	df1	df2	Sig. F Change
1	.578[a]	.335	.331	65.991	.335	99.587	1	198	<.001
2	.815[b]	.665	.660	47.087	.330	96.447	2	196	<.001

a. Predictors: (Constant), Advertsing budget (thousands)

b. Predictors: (Constant), Advertsing budget (thousands), Band image rating (0-10), No. of plays on radio

12. Model 1 accounts for 33.5% of the variance in album sales.
13. Model 2 incorporates two additional variables: the number of radio plays (Airplay) and the band's attractiveness (Image). This augmented model accounts for an *additional* 33% of the variance, a statistically significant increase ($\Delta R^2 = .33$, $p < .001$). Thus, Model 2 collectively explains 66.5% of the variance in album sales.

ANOVA[a]

Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	433687.833	1	433687.833	99.587	<.001[b]
	Residual	862264.167	198	4354.870		
	Total	1295952.000	199			
2	Regression	861377.418	3	287125.806	129.498	<.001[c]
	Residual	434574.582	196	2217.217		
	Total	1295952.000	199			

a. Dependent Variable: Album sales (thousands)

b. Predictors: (Constant), Advertsing budget (thousands)

c. Predictors: (Constant), Advertsing budget (thousands), Band image rating (0-10), No. of plays on radio

14. The **Model Summary** and **ANOVA** tables demonstrate that Model 2, *overall*, significantly predicts the outcome variable ($F(3, 196) = 129.50$, $p < .001$).

Coefficients[a]

Model		Unstandardized Coefficients		Standardized Coefficients	t	Sig.
		B	Std. Error	Beta		
1	(Constant)	134.140	7.537		17.799	<.001
	Advertsing budget (thousands)	.096	.010	.578	9.979	<.001
2	(Constant)	-26.613	17.350		-1.534	.127
	Advertsing budget (thousands)	.085	.007	.511	12.261	<.001
	No. of plays on radio	3.367	.278	.512	12.123	<.001
	Band image rating (0-10)	11.086	2.438	.192	4.548	<.001

a. Dependent Variable: Album sales (thousands)

15. The **Coefficients** table can be written into the following formula:

$$\text{Sales}_i = -26.613 + 0.085(\text{Adverts}_i) + 3.367(\text{Airplay}_i) + 11.086(\text{Image}_i)$$

16. The advertising budget, the number of radio plays, and the band's attractiveness are all significant predictors of album sales (all $p < .001$).

17. For example, when no money is spent on advertising (Adverts = 0), the album gets played on the radio 10 times (Airplay = 10), and the band is moderately attractive (Image = 5), sales are expected to be 62,487 albums.

$$\text{Sales}_i = -26.613 + 0.085(0) + 3.367(10) + 11.086(5)$$
$$\text{Sales}_i = 62.487$$

End of Demonstration

Practice

1. Watch the video <stats-multiple_regression.mp4> (26:00)

End of Practice

Independent *t*-test

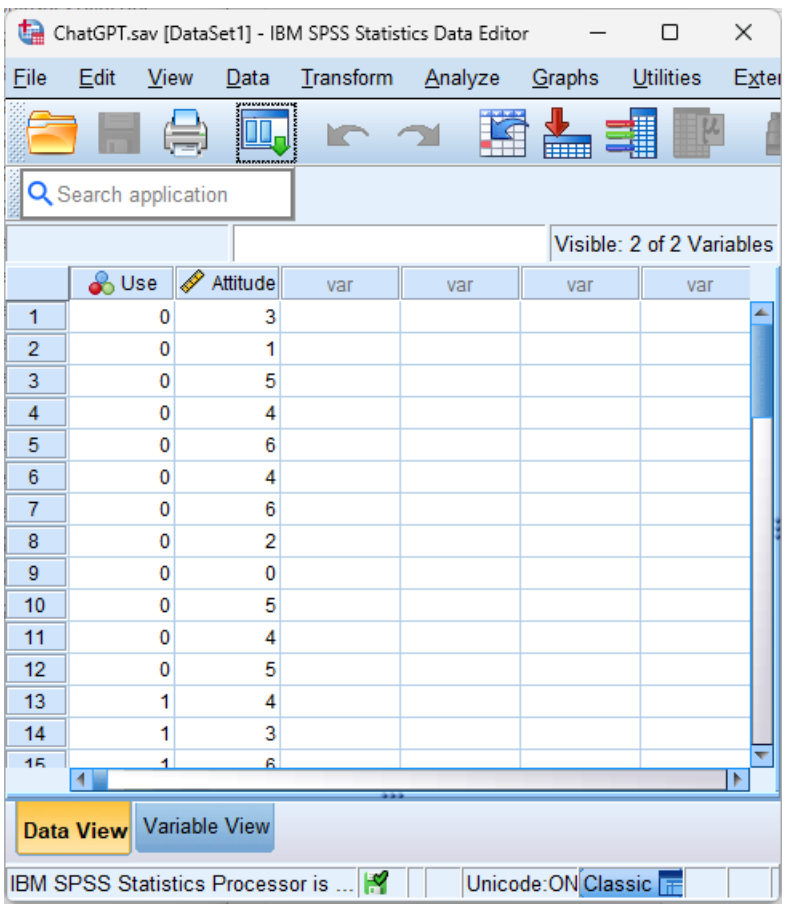
Goal: Run independent samples *t*-test and interpret the results.

OVERVIEW

1. Watch the video <stats-t_test.mp4> (12:25)
2. The three types of *t*-test include:
    a. **Independent samples *t*-test** compares the means of *two groups*.
    b. **Paired samples *t*-test** compares the means from the *same group* at different points in time (pre vs. post) or for different measures (variable A vs. variable B).
    c. **One sample *t*-test** compares the mean of a *single group against a known mean value*.

OPEN SYNTAX FILE AND DATA FILE

3. Open the "stats_soft-spss-inter_stats.sps" syntax file.
4. **File** >> **Open** >> **Data** >> Go to the directory containing the "ChatGPT.sav" data file and select the file >> **Paste** >> **Save** and execute the generated syntax.



a. A psychologist is investigating people's perspectives on using ChatGPT in their workplace.

b. She recruited 24 participants and asked whether they have ever utilized ChatGPT.

c. She also evaluated their stance on using ChatGPT in a work setting.

d. The Data View window displays one column for ChatGPT usage experience (Use: 0 = No, 1 = Yes) and another for the attitude toward using ChatGPT in the workplace (Attitude: 0 [negative] to 10 [positive]).

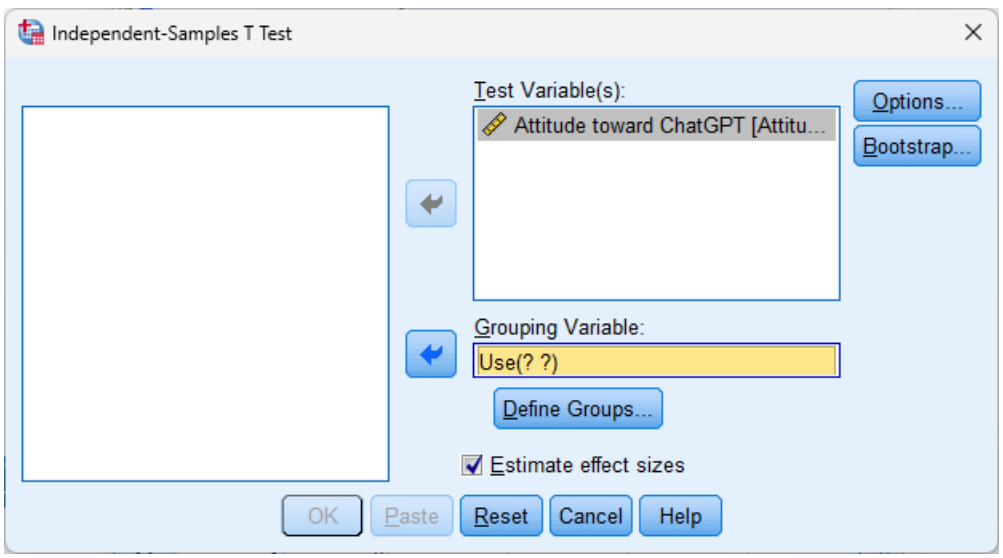
CONDUCT INDEPENDENT *T*-TEST

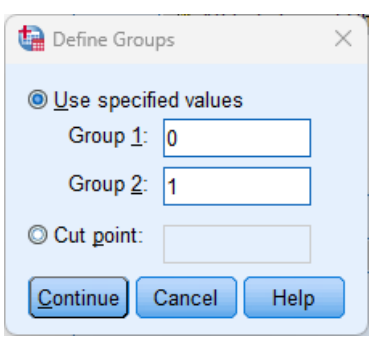
5. **Analyze** >> **Compare Means and Proportions** >> **Independent-Samples T Test** >> This will open the Independent-Samples T Test window.

6. In the left box of the window, select the variable Attitude >> Move this variable into the right Test Variable(s) box.

7. Select Use and move this variable into the Group Variable box >> Click **Define Groups**. This will open the Define Groups window.

8. In the Define Groups window, type "0" (No) into the Group 1 text box and "1" (Yes) into the Group 2 text box >> **Continue**.





9. **Paste** >> **Save** and execute the generated syntax.

INTERPRET RESULTS

Group Statistics

	Experience of ChatGPT	N	Mean	Std. Deviation	Std. Error Mean
Attitude toward ChatGPT	No	12	3.75	1.913	.552
	Yes	12	5.00	1.651	.477

Independent Samples Test

		Levene's Test for Equality of Variances		t-test for Equality of Means							
						Significance				95% Confidence Interval of the Difference	
		F	Sig.	t	df	One-Sided p	Two-Sided p	Mean Difference	Std. Error Difference	Lower	Upper
Attitude toward ChatGPT	Equal variances assumed	.545	.468	-1.713	22	.050	.101	-1.250	.730	-2.763	.263
	Equal variances not assumed			-1.713	21.541	.050	.101	-1.250	.730	-2.765	.265

Independent Samples Effect Sizes

		Standardizer[a]	Point Estimate	95% Confidence Interval	
				Lower	Upper
Attitude toward ChatGPT	Cohen's d	1.787	-.700	-1.518	.134
	Hedges' correction	1.851	-.675	-1.466	.129
	Glass's delta	1.651	-.757	-1.601	.117

a. The denominator used in estimating the effect sizes.
Cohen's d uses the pooled standard deviation.
Hedges' correction uses the pooled standard deviation, plus a correction factor.
Glass's delta uses the sample standard deviation of the control group.

10. *Equal variances assumed vs. Equal variances not assumed*: Parametric tests assume roughly equal variances in experimental groups. Levene's test examines the hypothesis of equal variances. A significant result ($p < .05$) of Levene's test suggests a violation of the assumption of homogeneous variances. Conversely, a non-significant result ($p \geq .05$) indicates that the variances are roughly equal and the assumption is valid.

11. For the current data, the Levene's test result is not significant ($p = .47$), allowing us to read the *t*-test statistics from the row labelled "Equal variances assumed". In cases where the Levene's test result is significant, the *t*-test statistics would be read from the row labelled "Equal variances not assumed".

12. On average, participants who have used ChatGPT exhibited a more positive attitude toward using ChatGPT in the workplace ($M = 5.00$, $SD = 1.65$) compared to those who had not used it before ($M = 3.75$, $SD = 1.91$). Although the difference in attitude scores between the two groups was large (Cohen's $d = 0.70$), it was not statistically significant, $t(22) = 1.71$, $p = .10$.

13. *Confidence intervals* are constructed to encompass the true value of the mean difference in 95% of random samples. Assuming this sample is among those containing the population value, we can infer that the *true* mean difference lies between –2.76 and 0.26. Notably, the intervals include 0, suggesting that the true value of the mean difference is likely to be 0.

14. ***[Quiz]*** Why does the instructor omit the minus sign when reporting the *d* and *t* values?

Practice

1. Watch the video <stats-central_limit_theorem.mp4> (7:34).
2. Watch the video <stats-p_values.mp4> (25:14).

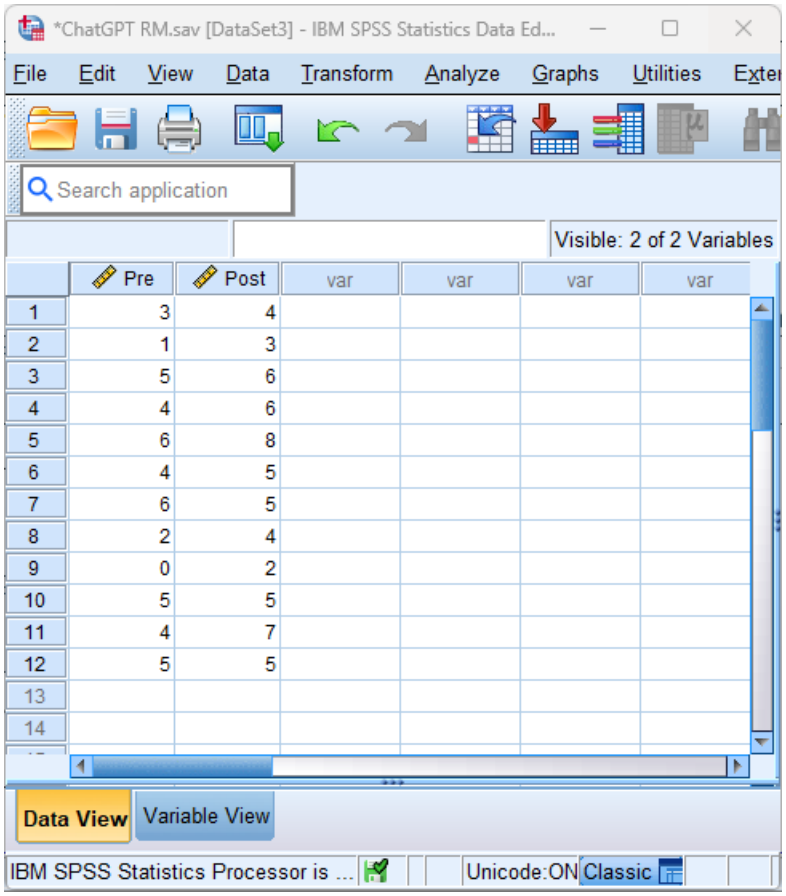
End of Practice

Paired *t*-test

Goal: Run paired samples *t*-test and interpret the results.

OPEN SYNTAX FILE AND DATA FILE

1. Open the "stats_soft-spss-inter_stats.sps" syntax file.
2. **File** >> **Open** >> **Data** >> Go to the directory containing the "ChatGPT RM.sav" data file and select the file >> **Paste** >> **Save** and execute the generated syntax.



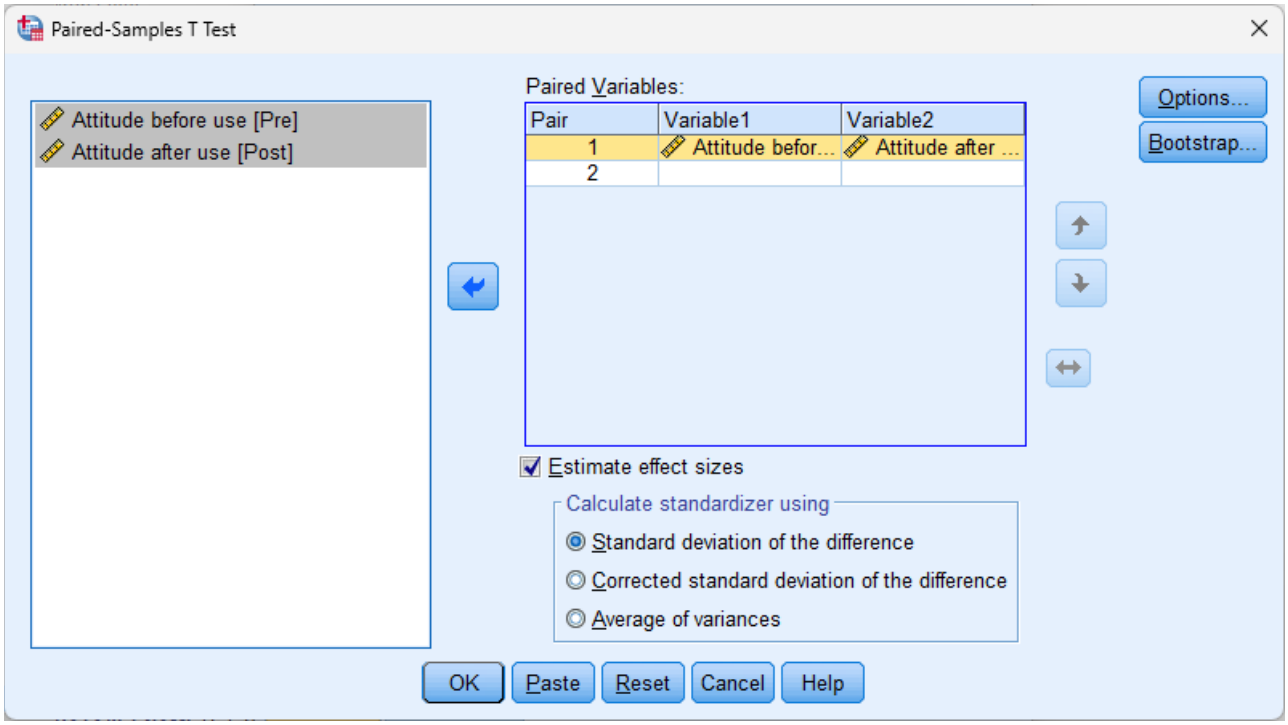
   a. The psychologist instructed the 12 participants who had never used ChatGPT to utilize the tool for a week. Following this period, she re-evaluated their attitude toward using ChatGPT in the workplace, employing a pre-post design.

   b. The Data View window displays two columns: the attitude toward using ChatGPT, measured before use (Pre) and after use (Post).

CONDUCT PAIRED *T*-TEST

3. **Analyze** >> **Compare Means and Proportions** >> **Paired-Samples T Test** >> This will open the Paired-Samples T Test window.

4. In the left box of the window, select the variables Pre and Post >> Move these variables into the right Paired Variables box.



5. **Paste** >> **Save** and executive the generated syntax.

INTERPRET RESULTS

Paired Samples Statistics

		Mean	N	Std. Deviation	Std. Error Mean
Pair 1	Attitude before use	3.75	12	1.913	.552
	Attitude after use	5.00	12	1.651	.477

Paired Samples Test

		Paired Differences					t	df	Significance	
					95% Confidence Interval of the Difference					
		Mean	Std. Deviation	Std. Error Mean	Lower	Upper			One-Sided p	Two-Sided p
Pair 1	Attitude before use - Attitude after use	-1.250	1.138	.329	-1.973	-.527	-3.804	11	.001	.003

Paired Samples Effect Sizes

					95% Confidence Interval	
			Standardizer[a]	Point Estimate	Lower	Upper
Pair 1	Attitude before use - Attitude after use	Cohen's d	1.138	-1.098	-1.808	-.358
		Hedges' correction	1.224	-1.021	-1.682	-.333

a. The denominator used in estimating the effect sizes.
Cohen's d uses the sample standard deviation of the mean difference.
Hedges' correction uses the sample standard deviation of the mean difference, plus a correction factor.

6. After using ChatGPT for a week, participants exhibited a significantly more positive attitude toward using ChatGPT in the workplace (*M* = 5.00, *SD* = 1.65) compared to before using it (*M* =

3.75, *SD* = 1.91), $t(11) = 3.80$, $p < .01$. The difference in attitude scores was large (Hedges' g = 1.02).

7. The 95% confidence intervals of the mean difference do not contain 0 (both limits are negative), indicating that the true value of the mean difference is unlikely to be 0. In other words, there is an effect in the *population* reflecting a more positive attitude toward ChatGPT when individuals have the opportunity to use it.
8. Hedges' *g* and Cohen's *d* are very similar except when sample sizes are below 20, where Hedges' *g* performs better. Hedges' *g* is often referred to as *corrected effect size*. For very small sample sizes (< 20), it is recommended to choose Hedge's *g* over Cohen's *d*.

End of Demonstration

Practice

1. Watch the video <stats-confidence_intervals.mp4> (10:55).

End of Practice

Analysis of variance (ANOVA)

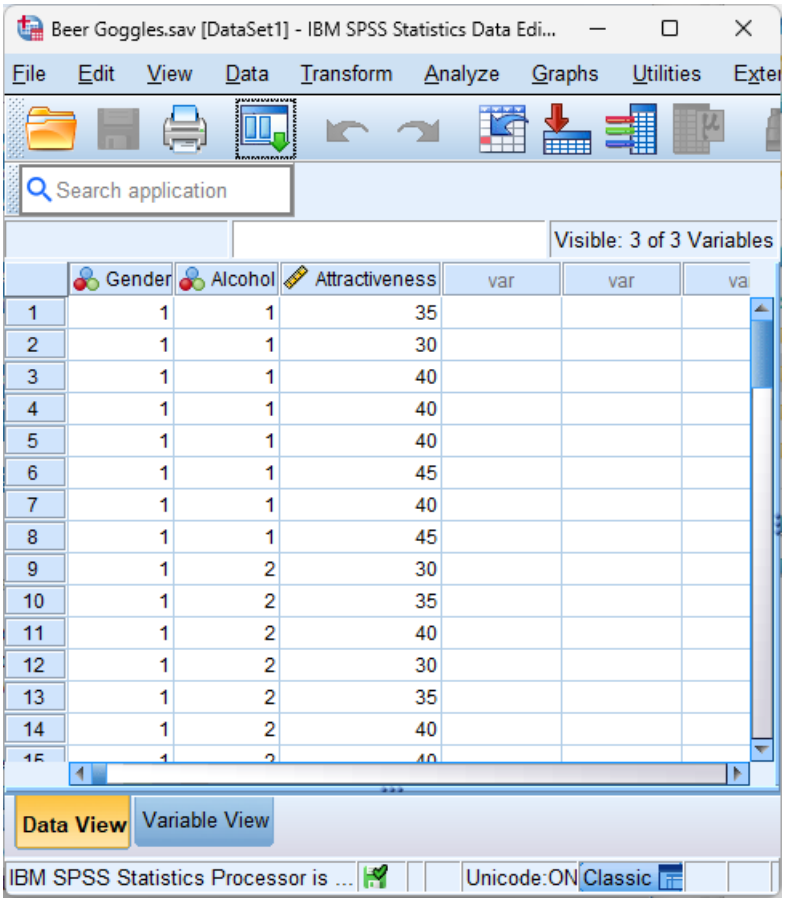
Goal: Run 2-way independent ANOVA and interpret the results.

OVERVIEW

1. Watch the video <stats-anova.mp4> (14:13)
2. When examining differences between group means, this scenario can be viewed as a special case of the linear model (linear regression) with a categorical predictor variable(s).

OPEN SYNTAX FILE AND DATA FILE

3. Open the "stats_soft-spss-inter_stats.sps" syntax file.
4. **File** >> **Open** >> **Data** >> Go to the directory containing the "Beer Goggles.sav" data file and select the file >> **Paste** >> **Save** and execute the generated syntax.



	Gender	Alcohol	Attractiveness	var	var	va
1	1	1	35			
2	1	1	30			
3	1	1	40			
4	1	1	40			
5	1	1	40			
6	1	1	45			
7	1	1	40			
8	1	1	45			
9	1	2	30			
10	1	2	35			
11	1	2	40			
12	1	2	30			
13	1	2	35			
14	1	2	40			

   a. A psychologist is conducting a study on the influence of alcohol consumption on subjective perception. Her hypothesis is that after consuming alcohol, subjective perception of physical attractiveness would become more distorted, commonly referred to as the 'beer-goggles effect.'

   b. Additionally, she aims to investigate whether this effect differs between men and women.

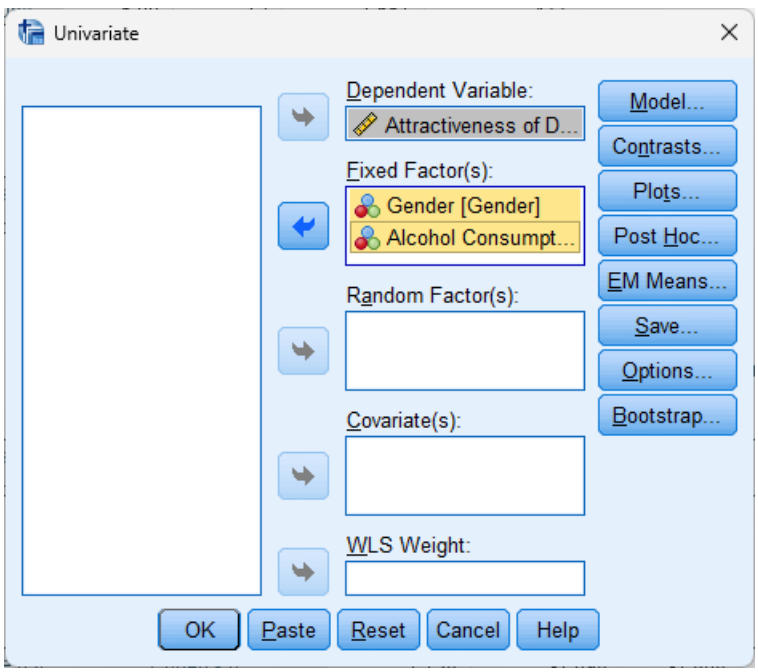
c. For the study, 48 college students were recruited, evenly divided into three groups: no alcohol, 2 pints of beer, or 4 pints of beer. Each group consisted of 16 participants, half of whom were males and the other half were females.

d. Each participant was asked to rate the attractiveness of 20 people provided on photographs.

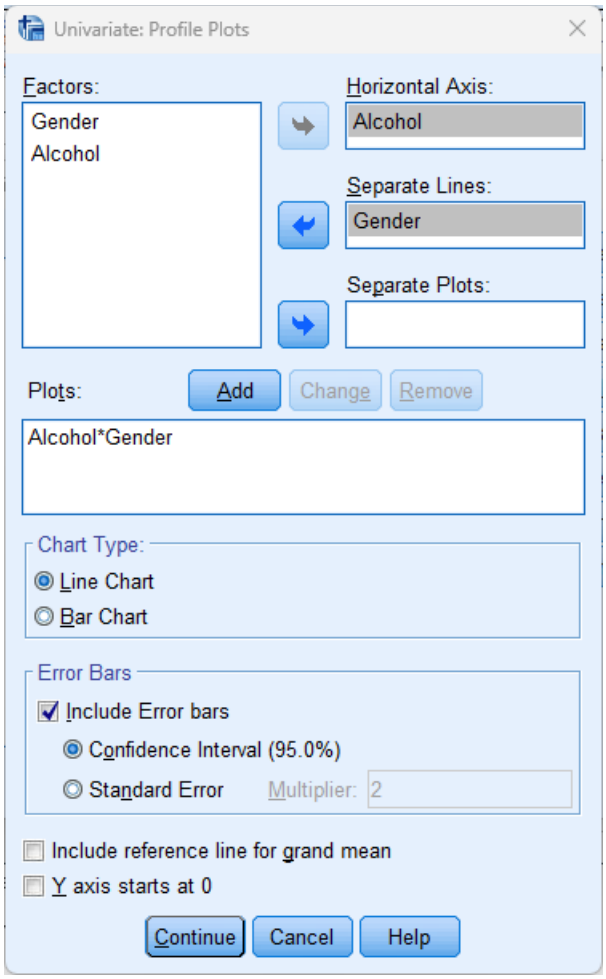
e. The Data View window displays three columns: Gender of the participant (0 = male, 1 = female), the amount of Alcohol consumed (1 = no alcohol, 2 = 2 pints, 3 = 3 pints), and the average rating of Attractiveness (1 to 100).

CONDUCT ANOVA

5. **Analyze** >> **General Linear Model** >> **Univariate** >> This will open the Univariate window.
6. In the left box of the window, select the variable Attractiveness >> Move this variable to the right Dependent Variable box.
7. Select Gender and Alcohol and move these variables into the Fixed Factor(s) box.

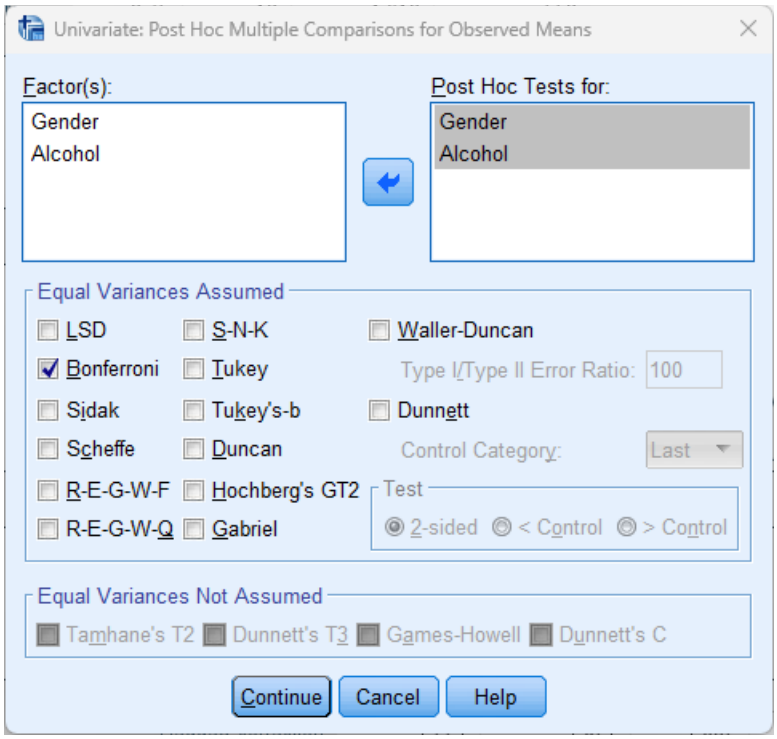


8. Plotting is a useful tool for visualizing/confirming *interactions* between the variables. **Plots** >> Move the Alcohol variable from the Factors box into the Horizonal Axis box >> Similarly, move the Gender variable into the Separate Lines box >> **Add** >> Select **Line Chart** and **Include Error bars** options >> **Continue**.

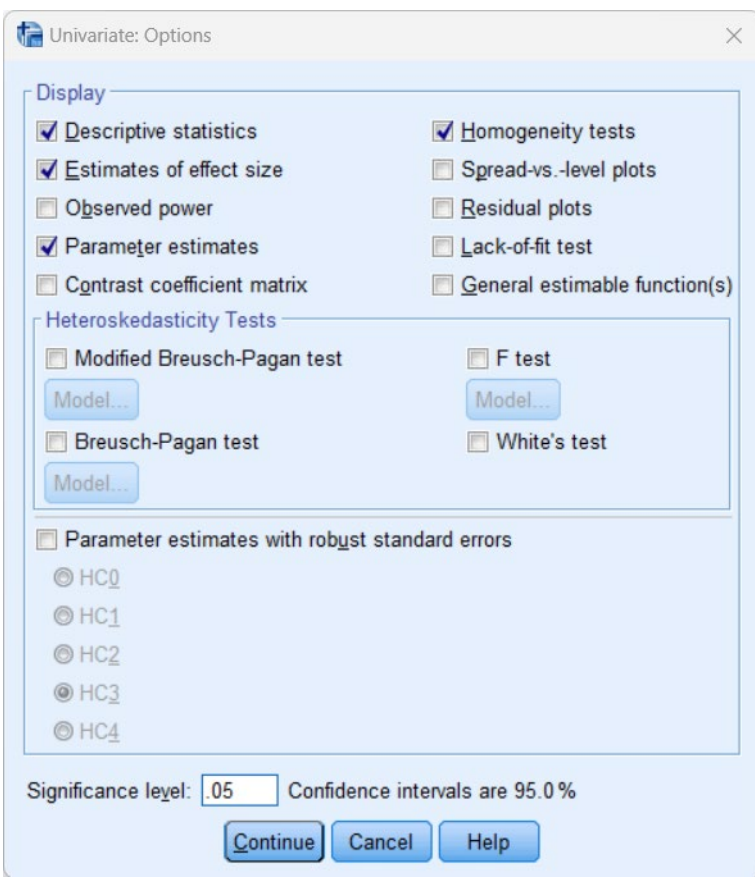


9. **Post Hoc** >> Select Gender and Alcohol in the Factor(s) box and move these variables into the Post Hoc Tests for box >> Select **Bonferroni** option >> **Continue**.

10. ***[Note]*** Post-hoc tests, known as 'pairwise' comparisons, will not be conducted for Gender because there are fewer than three groups. SPSS will issue a warning message and ignore post-hoc comparisons for Gender (although it will perform post-hoc comparisons for Alcohol).



11. **Options** >> Select options >> **Continue**.



12. **Paste** >> **Save** and run the generated syntax.

INTERPRET RESULTS

Levene's Test of Equality of Error Variances[a,b]

		Levene Statistic	df1	df2	Sig.
Attractiveness of Date	Based on Mean	1.527	5	42	.202
	Based on Median	1.425	5	42	.235
	Based on Median and with adjusted df	1.425	5	33.323	.241
	Based on trimmed mean	1.536	5	42	.199

Tests the null hypothesis that the error variance of the dependent variable is equal across groups.

a. Dependent variable: Attractiveness of Date

b. Design: Intercept + Gender + Alcohol + Gender * Alcohol

13. The result of Levene's test is ***not significant***, indicating that the assumption of homogeneous variances is met.

Tests of Between-Subjects Effects

Dependent Variable: Attractiveness of Date

Source	Type III Sum of Squares	df	Mean Square	F	Sig.	Partial Eta Squared
Corrected Model	5479.167[a]	5	1095.833	13.197	<.001	.611
Intercept	163333.333	1	163333.333	1967.025	<.001	.979
Gender	168.750	1	168.750	2.032	.161	.046
Alcohol	3332.292	2	1666.146	20.065	<.001	.489
Gender * Alcohol	1978.125	2	989.062	11.911	<.001	.362
Error	3487.500	42	83.036			
Total	172300.000	48				
Corrected Total	8966.667	47				

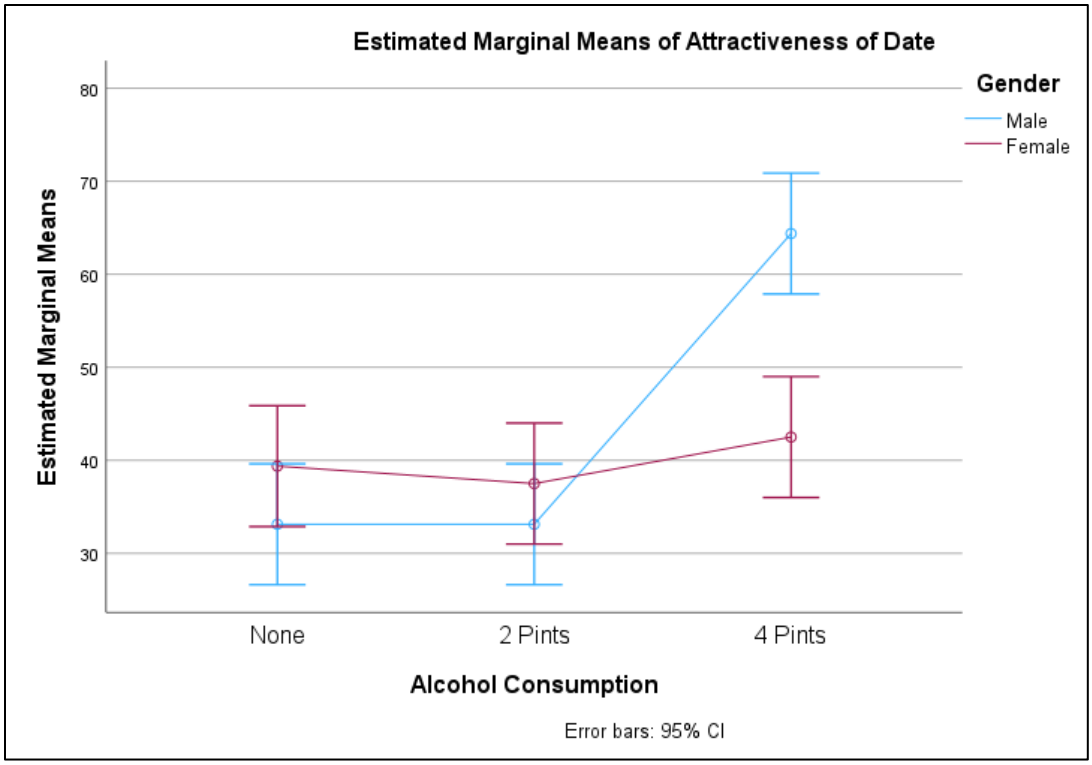
a. R Squared = .611 (Adjusted R Squared = .565)

14. The main effect of alcohol consumption on the perception of physical attractiveness is significant, $F(2, 42) = 20.07$, $p < .001$, partial $\eta^2 = .49$. This finding indicates that, overall, disregarding the gender of the participants, the amount of alcohol consumed influenced their ratings of attractiveness in others.

15. Post-hoc pairwise comparisons reveal that participants who consumed no alcohol or had 2 pints of beer provided similar ratings of attractiveness (adjusted $p = 1.00$). However, the group that consumed 4 pints of beer gave significantly higher ratings compared to the other two groups (both adjusted $p < .001$).

16. The main effect of gender on the perception of attractiveness is not significant, $F(1, 42) = 2.03$, $p = .16$, partial $\eta^2 = .05$. This finding indicates that, overall, regardless of the amount of alcohol consumed, the gender of the participants did not influence their ratings of attractiveness in others.

17. Importantly, there is a significant interaction between alcohol consumption and gender, $F(2, 42) = 11.91$, $p < .001$, partial $\eta^2 = .36$. This finding suggests that the impact of alcohol consumption on the perception of attractiveness differed between male and female participants.



18. The graph of *estimated marginal means* provides insights into the nature of this interaction. Specifically, for women, alcohol consumption had minimal influence as the attractiveness ratings were relatively stable across the three alcohol groups. In contrast, for men, the attractiveness ratings were stable with a small amount of alcohol but rapidly increased when a larger amount was consumed (i.e., 4 pints).
19. Significant interactions are often confirmed by non-parallel lines or lines that intersect.
20. To conclude, alcohol consumption can lead to distorted perceptions of physical attractiveness. However, such 'beer-goggles effect' is particularly pronounced among males. This serves as an example of how main effects can be misleading when a significant interaction(s) is present!

Parameter Estimates

Dependent Variable: Attractiveness of Date

Parameter	B	Std. Error	t	Sig.	95% Confidence Interval Lower Bound	95% Confidence Interval Upper Bound	Partial Eta Squared
Intercept	42.500	3.222	13.192	<.001	35.998	49.002	.806
[Gender=0]	21.875	4.556	4.801	<.001	12.680	31.070	.354
[Gender=1]	0[a]	.	.	.	.	.	.
[Alcohol=1]	-3.125	4.556	-.686	.497	-12.320	6.070	.011
[Alcohol=2]	-5.000	4.556	-1.097	.279	-14.195	4.195	.028
[Alcohol=3]	0[a]	.	.	.	.	.	.
[Gender=0] * [Alcohol=1]	-28.125	6.443	-4.365	<.001	-41.128	-15.122	.312
[Gender=0] * [Alcohol=2]	-26.250	6.443	-4.074	<.001	-39.253	-13.247	.283
[Gender=0] * [Alcohol=3]	0[a]	.	.	.	.	.	.
[Gender=1] * [Alcohol=1]	0[a]	.	.	.	.	.	.
[Gender=1] * [Alcohol=2]	0[a]	.	.	.	.	.	.
[Gender=1] * [Alcohol=3]	0[a]	.	.	.	.	.	.

a. This parameter is set to zero because it is redundant.

21. Again, ANOVA can be viewed as a special case of the linear model (linear regression) with a categorical predictor variable(s). The **Parameter Estimates** table expresses the ANOVA results as in the linear modeling results (i.e., regression coefficients):

$$\text{Attractiveness}_i = 42.50 + 21.88(\text{Male}_i) - 3.13(\text{No alcohol}_i) - 5.00(2\text{ Pints}_i) - 28.13(\text{Male}_i * \text{No alcohol}_i) - 26.25(\text{Male}_i * 2\text{ Pints}_i)$$

For females ($\text{Male}_i = 0$), this equation is simplified to:

$$\text{Attractiveness}_i = 42.50 - 3.13(\text{No alcohol}_i) - 5.00(2\text{ Pints}_i)$$

a. If females had no alcohol, 42.50 - 3.13(1) - 5.00(0) = 39.38.
b. If females drank 2 pints of beer, 42.50 - 3.13(0) - 5.00(1) = 37.50.
c. If females drank 4 pints of beer, 42.50 - 3.13(0) - 5.00(0) = 42.50.

For males ($\text{Male}_i = 1$), this equation is simplified to:

$$\text{Attractiveness}_i = 64.38 - 31.25(\text{No alcohol}_i) - 31.25(2\text{ Pints}_i)$$

a. If males had no alcohol, 64.38 - 31.25(1) - 31.25(0) = 33.13.
b. If males drank 2 pints of beer, 64.38 - 31.25(0) - 31.25(1) = 33.13.
c. If males drank 4 pints of beer, 64.38 - 31.25(0) - 31.25(0) = 64.38.

End of Demonstration

Practice

1. Read the article <nordstokke_zumbo-2007>: A cautionary tale about Levene's tests for equal variances.

End of Practice

Analysis of covariance (ANCOVA)

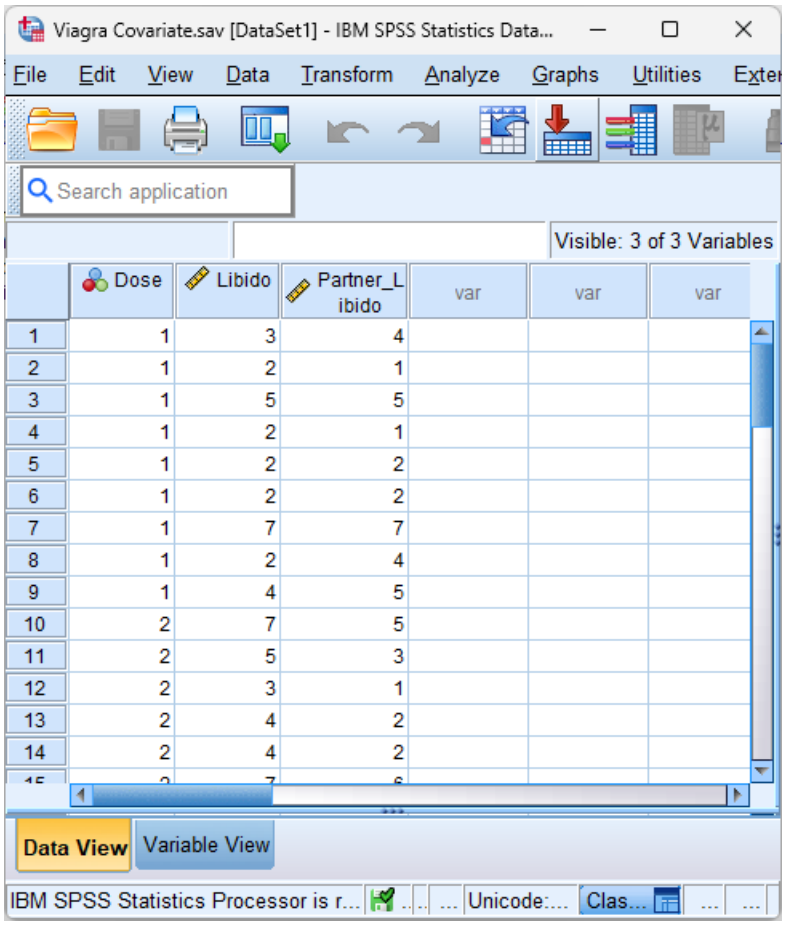
Goal: Run 2-way independent ANOVA and interpret the results.

OVERVIEW

1. Watch the video <stats-ancova.mp4> (13:10)
2. ANOVA can be expanded by incorporating one or more *continuous* variables that predict the outcome variable.
3. Continuous variables, such as those that are unrelated to the main experimental manipulation but still influence the outcome variable, are referred to as *covariates*.
4. The inclusion of covariates in ANOVA serves two purposes:
    a. *Reduction of within-group error variance*: In ANOVA, we evaluate the effect of an experiment by comparing the extent to which the experiment explains variability in the data against the remaining unexplained variability. By accounting for some of this 'unexplained' variance through covariates, we can diminish error variance and more accurately assess the effect of the experimental manipulation.
    b. *Elimination of confounding factors*: In any experiment, there may be unmeasured variables that systematically vary along with the experiment manipulation, leading to potential confounding effects. Once a potential confounding variable is identified, it can be measured and included as a covariate in the analysis, thereby mitigating its impact on the results.

OPEN SYNTAX FILE AND DATA FILE

5. Open the "stats_soft-spss-inter_stats.sps" syntax file.
6. **File** >> **Open** >> **Data** >> Go to the directory containing the "Viagra Covariate.sav" data file and select the file >> **Paste** >> **Save** and execute the generated syntax.



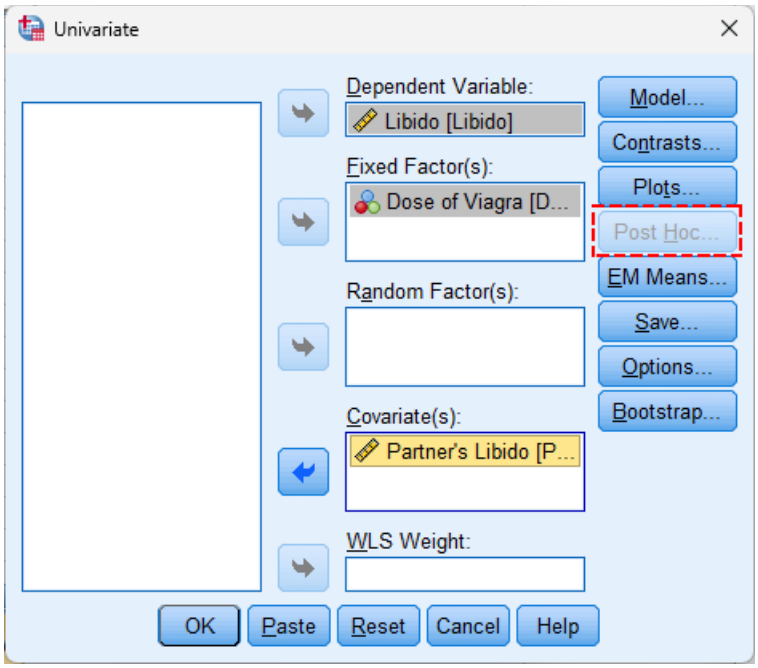
	Dose	Libido	Partner_Libido	var	var	var
1	1	3	4			
2	1	2	1			
3	1	5	5			
4	1	2	1			
5	1	2	2			
6	1	2	2			
7	1	7	7			
8	1	2	4			
9	1	4	5			
10	2	7	5			
11	2	5	3			
12	2	3	1			
13	2	4	2			
14	2	4	2			

a. In the psychology literature, sexual performance issues are often linked to a decrease in libido (Hawton, 1989). To investigate this relationship, a psychologist is conducting a study involving three Dose conditions. The first group received a placebo, the second group received a low dose of Viagra (a sexual stimulant), and the third group received a high dose of Viagra. An objective measure of libido (Libido) was used as the outcome variable.

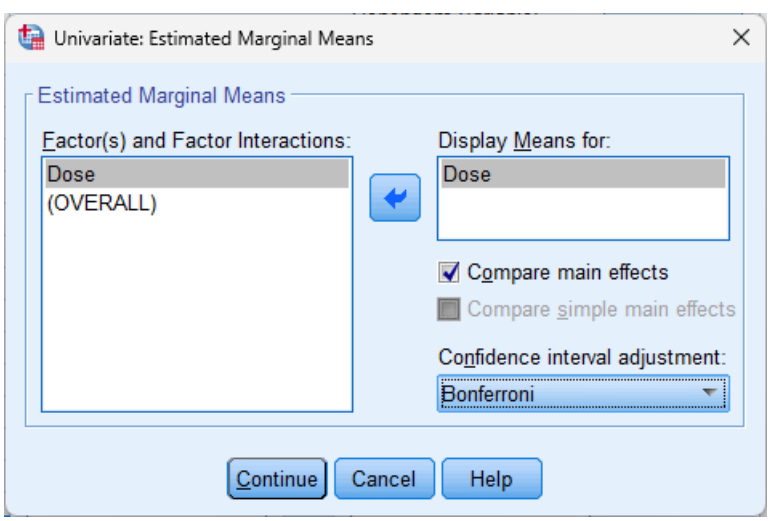
b. In addition to examining the participant's own libido, the psychologist hypothesized that the libido of their sexual partners would also have an influence. Hence, she measured the partner's libido, specifically assessing how frequently they initiated sexual contact (Partner_Libido).

CONDUCT ANCOVA

22. **Analyze** >> **General Linear Model** >> **Univariate** >> This will open the Univariate window.
23. In the left box of the window, select the variable Libido >> Move this variable into the right Dependent Variable box.
24. Select Dose and move this variable into the Fixed Factor(s) box.
25. Select Partner_Libido and move this variable into the Covariate(s) box.



26. When a covariate is included, post-hoc tests are not applicable. Post-hoc comparisons of the 'original' group means are not appropriate in the presence of a covariate because *group means that have been adjusted for the covariate* should be used for interpretation.
27. To conduct post-hoc pairwise comparisons of "estimated marginal means", **EM Means** >> Select Dose from the Factor(s) and Factor Interactions box and move this variable into the Display Means for box >> Select **Compare main effects** option >> Select **Bonferroni** from the Confidence interval adjustment drop-down menu >> **Continue**.
28. It is not recommended to use the **LSD(non)** option, which applies Tukey LSD test without any adjustment. If you are concerned about the potential loss of power associated with Bonferroni correction being overly conservative, it is advisable to select the **Sidak** option instead.



29. **Options** >> Select options >> **Continue**.
30. **Paste** >> **Save** and run the generated syntax.

INTERPRET RESULTS

Tests of Between-Subjects Effects

Dependent Variable: Libido

Source	Type III Sum of Squares	df	Mean Square	F	Sig.
Corrected Model	16.844[a]	2	8.422	2.416	.108
Intercept	535.184	1	535.184	153.522	<.001
Dose	16.844	2	8.422	2.416	.108
Error	94.123	27	3.486		
Total	683.000	30			
Corrected Total	110.967	29			

a. R Squared = .152 (Adjusted R Squared = .089)

31. This table presents the ANOVA results *when the covariate is not included*. The administration of Viagra does not have a significant impact on libido.

Tests of Between-Subjects Effects

Dependent Variable: Libido

Source	Type III Sum of Squares	df	Mean Square	F	Sig.	Partial Eta Squared
Corrected Model	31.920[a]	3	10.640	3.500	.030	.288
Intercept	76.069	1	76.069	25.020	<.001	.490
Partner_Libido	15.076	1	15.076	4.959	.035	.160
Dose	25.185	2	12.593	4.142	.027	.242
Error	79.047	26	3.040			
Total	683.000	30				
Corrected Total	110.967	29				

a. R Squared = .288 (Adjusted R Squared = .205)

32. In contrast, this table displays the ANOVA results *when the covariate is included*. Controlling for the partner's libido, the effect of Viagra on libido is significant, $F(2, 26) = 4.14$, $p < .05$, partial $\eta^2 = .24$.

Estimates

Dependent Variable: Libido

Dose of Viagra	Mean	Std. Error	95% Confidence Interval	
			Lower Bound	Upper Bound
Placebo	2.926[a]	.596	1.701	4.152
Low Dose	4.712[a]	.621	3.436	5.988
High Dose	5.151[a]	.503	4.118	6.184

a. Covariates appearing in the model are evaluated at the following values: Partner's Libido = 2.73.

Pairwise Comparisons

Dependent Variable: Libido

(I) Dose of Viagra	(J) Dose of Viagra	Mean Difference (I-J)	Std. Error	Sig.[b]	95% Confidence Interval for Difference[b] Lower Bound	Upper Bound
Placebo	Low Dose	-1.786	.849	.136	-3.959	.388
	High Dose	-2.225*	.803	.031	-4.279	-.171
Low Dose	Placebo	1.786	.849	.136	-.388	3.959
	High Dose	-.439	.811	1.000	-2.515	1.637
High Dose	Placebo	2.225*	.803	.031	.171	4.279
	Low Dose	.439	.811	1.000	-1.637	2.515

Based on estimated marginal means

*. The mean difference is significant at the .05 level.

b. Adjustment for multiple comparisons: Bonferroni.

33. The **Estimates** and **Pairwise Comparisons** tables provide adjusted group means, *accounting for the influence of the covariate*, and the results of pairwise comparisons of these means. The high-dose group (M = 5.15, SE = 0.50) exhibited significantly higher libidos compared to the placebo group (M = 2.93, SE = 0.60) (adjusted $p < .05$). However, there were no significant differences between the low-dose group (M = 4.71, SE = 0.62) and either the placebo group (adjusted p = .14) or the high-dose group (adjusted p = 1.00).

End of Demonstration

Repeated measures analysis of variance (RM ANOVA)

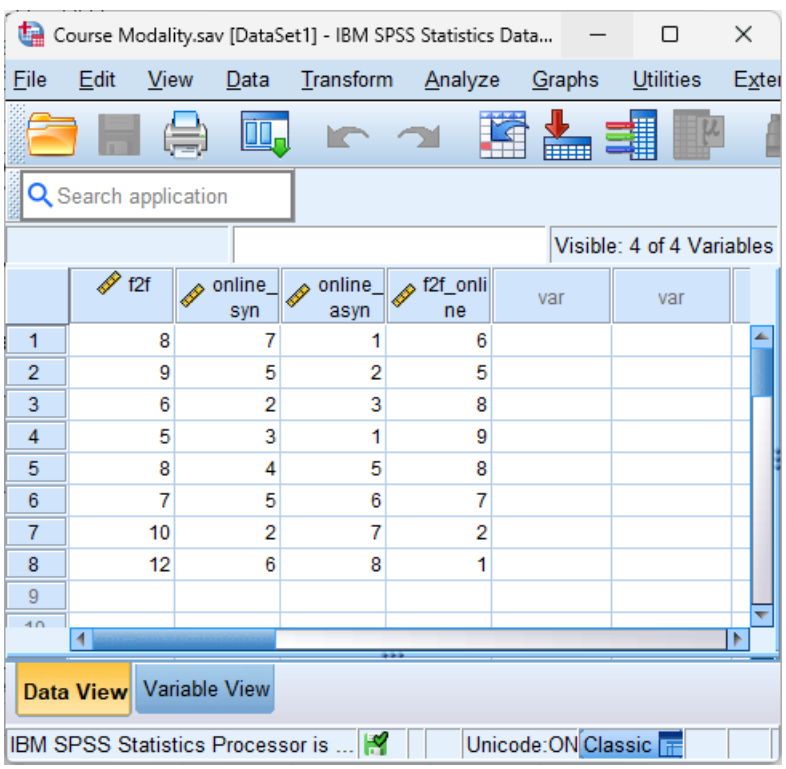
Goal: Run repeated measures ANOVA and interpret the results.

OVERVIEW

1. Watch the video <stats-rm_anova.mp4> (14:12)
2. Parametric tests typically assume that data points are *independent*. However, in a repeated measures design, this is not the case because the data for different conditions originates from the same entities (persons).
3. Given this, an additional assumption beyond those of independent ANOVA is necessary. In basic terms, we posit that the "relationship" between pairs of experimental conditions is similar. This assumption is referred to as *sphericity*.
4. *Mauchly's test* is designed to evaluate the hypothesis that the variances of differences between the conditions are equal.

OPEN SYNTAX FILE AND DATA FILE

5. Open the "stats_soft-spss-inter_stats.sps" syntax file.
6. **File** >> **Open** >> **Data** >> Go to the directory containing the "Course Modality.sav" data file and select the file >> **Paste** >> **Save** and execute the generated syntax.



	f2f	online_syn	online_asyn	f2f_online	var	var
1	8	7	1	6		
2	9	5	2	5		
3	6	2	3	8		
4	5	3	1	9		
5	8	4	5	8		
6	7	5	6	7		
7	10	2	7	2		
8	12	6	8	1		
9						

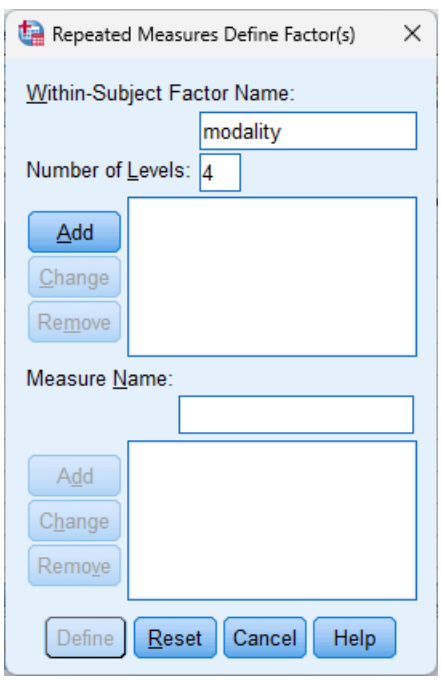
   a. An instructor is interested in understanding students' preference for the mode of course delivery. He provided four modules of a Survey Research course to eight graduate students.

b. Each of these four modules was presented in a different mode of delivery (one mode per module): face-to-face, synchronous online, asynchronous online, and a combination of face-to-face and asynchronous online. At the end of each module, students rated their satisfaction with the module on a scale of 1 to 10 (1 = very unsatisfactory, 10 = very satisfactory).

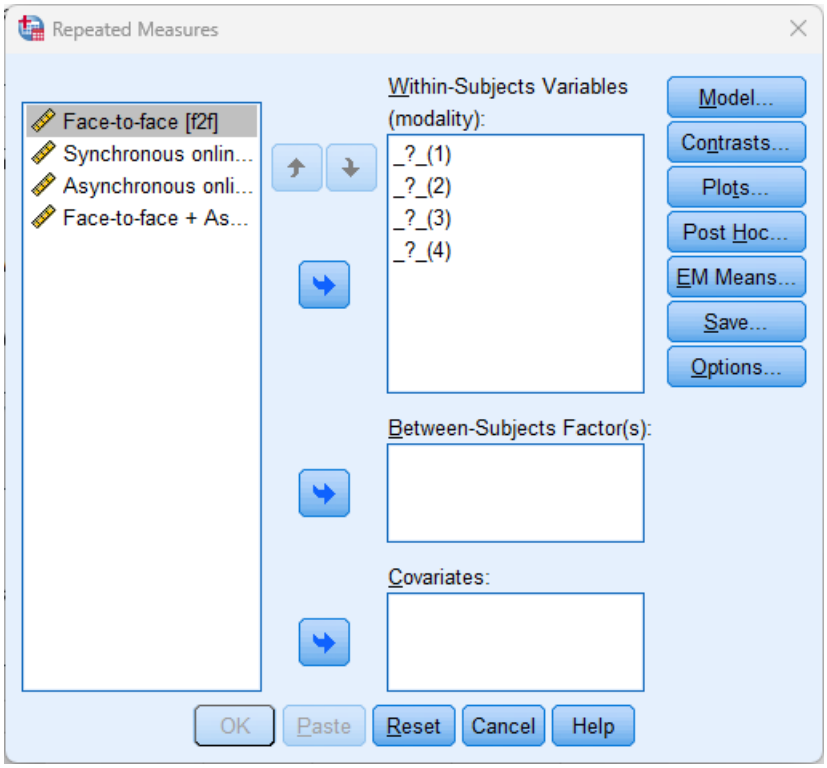
c. In this context, the mode of course delivery is the independent variable, while the students' preference is repeated measures, dependent variable. The Data View window displays four columns corresponding to the four different course delivery modes: f2f, online_syn, online_asyn, f2f_online.

CONDUCT RM ANOVA

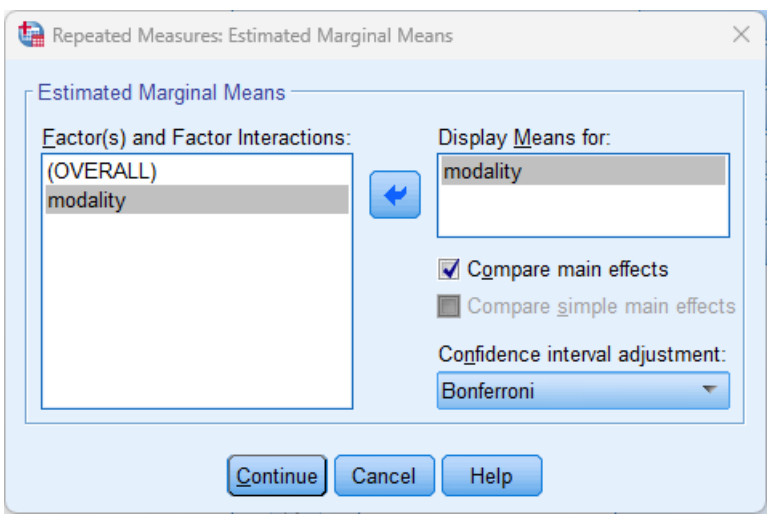
7. **Analyze** >> **General Linear Model** >> **Repeated Measures** >> This will open the Repeated Measures Define Factor(s) window.
8. Provide a name for the repeated measures (within-subject) variable. Replace "factor1" with "modality" in the Within-Subject Factor Name box.
9. Additionally, the number of levels for this variable (i.e., the number of experimental conditions) needs to be specified. Enter "4" into the Number of Levels box >> **Add** >> **Define**.



10. The Repeated Measures window has a box labelled Within-Subjects Variables, which displays a list of four question marks each followed by a number. These question marks represent placeholders for the columns that correspond to the four repeated measures. Select f2f, online_syn, online_asyn, and f2f_online and move these variables into the box.



11. To conduct post-hoc pairwise comparisons of "estimated marginal means", **EM Means** >> Select Modality from the Factor(s) and Factor Interactions box and move this variable into the Display Means for box >> Select **Compare main effects** option >> Select **Bonferroni** from the Confidence interval adjustment drop-down menu >> **Continue**.



12. **Options** >> Select options >> **Continue**.
13. **Paste** >> **Save** and run the generated syntax.

INTERPRET RESULTS

Mauchly's Test of Sphericity[a]

Measure: MEASURE_1

Within Subjects Effect	Mauchly's W	Approx. Chi-Square	df	Sig.	Epsilon[b] Greenhouse-Geisser	Huynh-Feldt	Lower-bound
modality	.136	11.406	5	.047	.533	.666	.333

Tests the null hypothesis that the error covariance matrix of the orthonormalized transformed dependent variables is proportional to an identity matrix.

a. Design: Intercept
Within Subjects Design: modality

b. May be used to adjust the degrees of freedom for the averaged tests of significance. Corrected tests are displayed in the Tests of Within-Subjects Effects table.

14. SPSS prints out the result of *Mauchly's test*, which assesses whether the data violates the assumption of sphericity.
    a. If Mauchly's test result is not significant (i.e., $p > .05$), the assumption of sphericity is met.
    b. If Mauchly's test result is significant (i.e., $p < .05$), the assumption of sphericity is not met.
        i. If the data violates the assumption of sphericity, we adjust the degrees of freedom (*df*) for the effect by multiplying it with one of the sphericity estimates ε. This will reduce the *df*, making the *F*-test more conservative (i.e., the effect should be sufficiently large to be deemed significant).
15. Mauchly's test indicates that the data violates the assumption of sphericity, $\chi^2(5) = 11.41$, $p < .05$.

Tests of Within-Subjects Effects

Measure: MEASURE_1

Source		Type III Sum of Squares	df	Mean Square	F	Sig.	Partial Eta Squared
modality	Sphericity Assumed	83.125	3	27.708	3.794	.026	.351
	Greenhouse-Geisser	83.125	1.599	52.001	3.794	.063	.351
	Huynh-Feldt	83.125	1.997	41.619	3.794	.048	.351
	Lower-bound	83.125	1.000	83.125	3.794	.092	.351
Error(modality)	Sphericity Assumed	153.375	21	7.304			
	Greenhouse-Geisser	153.375	11.190	13.707			
	Huynh-Feldt	153.375	13.981	10.970			
	Lower-bound	153.375	7.000	21.911			

16. *With Greenhouse-Geisser correction*: There is no significant difference in student satisfaction across the four course modalities, $F(1.60, 11.19) = 3.79$, $p = .06$, partial $\eta^2 = .35$.

    With Huynh-Feldt correction: There is significant difference in student satisfaction across the four course modalities, $F(2, 13.98) = 3.79$, $p < .05$, partial $\eta^2 = .35$.

    With lower bound correction: There is no significant difference in student satisfaction across the four course modalities, $F(1, 7) = 3.79$, p .09, partial $\eta^2 = .35$.

    (Note that while the *F* value and effect size remain unchanged, only the *df* and resulting *p*-value do alter)
17. This scenario highlights the arbitrariness of the .05 criterion for significance. Clearly, these *F* values represent the same-sized effect, but the effect is deemed significant under one criterion and not under another!

Estimates

Measure: MEASURE_1

modality	Mean	Std. Error	95% Confidence Interval Lower Bound	95% Confidence Interval Upper Bound
1	8.125	.789	6.259	9.991
2	4.250	.648	2.718	5.782
3	4.125	.972	1.827	6.423
4	5.750	1.031	3.313	8.187

Pairwise Comparisons

Measure: MEASURE_1

(I) modality	(J) modality	Mean Difference (I-J)	Std. Error	Sig.[b]	95% Confidence Interval for Difference[b] Lower Bound	95% Confidence Interval for Difference[b] Upper Bound
1	2	3.875*	.811	.012	.925	6.825
	3	4.000*	.732	.006	1.339	6.661
	4	2.375	1.792	1.000	-4.141	8.891
2	1	-3.875*	.811	.012	-6.825	-.925
	3	.125	1.202	1.000	-4.244	4.494
	4	-1.500	1.336	1.000	-6.359	3.359
3	1	-4.000*	.732	.006	-6.661	-1.339
	2	-.125	1.202	1.000	-4.494	4.244
	4	-1.625	1.822	1.000	-8.249	4.999
4	1	-2.375	1.792	1.000	-8.891	4.141
	2	1.500	1.336	1.000	-3.359	6.359
	3	1.625	1.822	1.000	-4.999	8.249

Based on estimated marginal means

*. The mean difference is significant at the .05 level.

b. Adjustment for multiple comparisons: Bonferroni.

18. The **Estimates** and **Pairwise Comparisons** tables provide adjusted group means, and the results of pairwise comparisons of these means. Students reported significantly higher satisfaction with the f2f module (M = 8.13, SE = 0.79) compared to both the synchronous online (M = 4.25, SE = 0.65, adjusted $p < .05$) and asynchronous online (M = 4.13, SE = 0.97, adjusted $p < .01$) modules.

19. *How to report*: Mauchly's test indicated the assumption of sphericity had been violated, $\chi^2(5) = 11.41$, $p < .05$, therefore the degrees of freedom were corrected using Huynh-Feldt estimate of sphericity ($\varepsilon = 0.67$). The results show that there was a significant effect of course delivery mode on student satisfaction, $F(2.00, 13.98) = 3.79$, $p < .05$, partial $\eta^2 = .35$. Post-hoc pairwise comparisons at a Bonferroni corrected alpha level further revealed that students preferred a face-to-face module (M = 8.13, SE = 0.79) more than an online module either synchronous (M = 4.25, SE = 0.65, adjusted $p < .05$) or asynchronous (M = 4.13, SE = 0.97, adjusted $p < .01$).

End of Demonstration

Mixed analysis of variance (Mixed ANOVA)

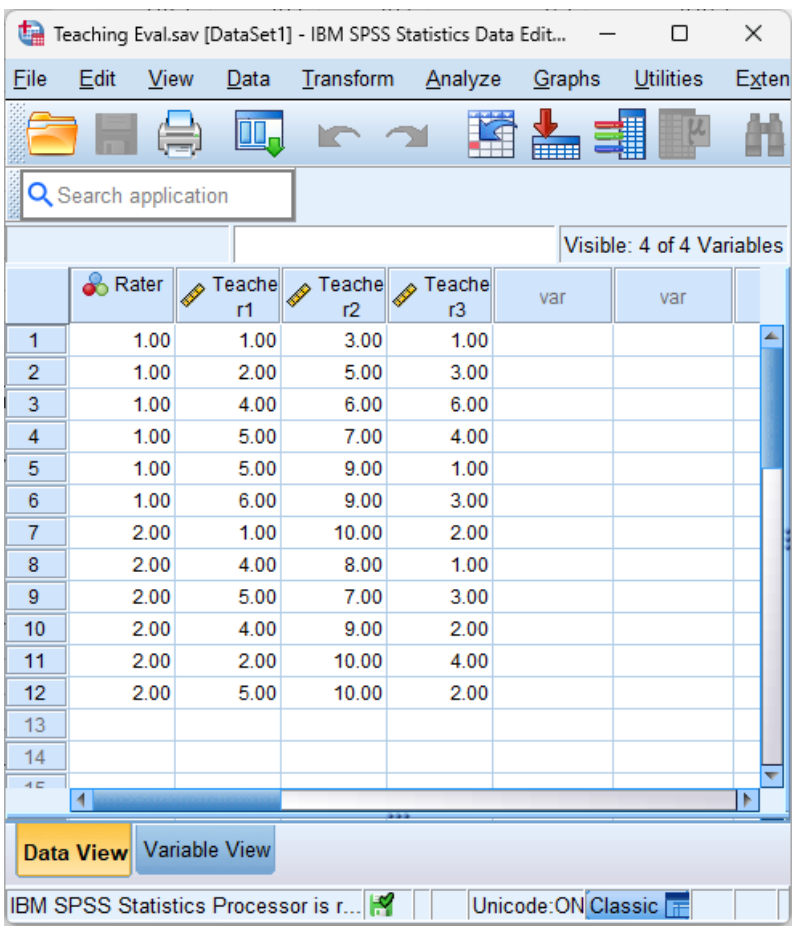
Goal: Run mixed ANOVA and interpret the results.

OVERVIEW

1. Watch the video <stats-mixed_anova.mp4> (11:31)
2. The final ANOVA design we will explore involves a combination of between-group variable(s) and repeated measures variable(s).

OPEN SYNTAX FILE AND DATA FILE

3. Open the "stats_soft-spss-inter_stats.sps" syntax file.
4. **File** >> **Open** >> **Data** >> Go to the directory containing the "Teaching Eval.sav" data file and select the file >> **Paste** >> **Save** and execute the generated syntax.



	Rater	Teacher1	Teacher2	Teacher3	var	var
1	1.00	1.00	3.00	1.00		
2	1.00	2.00	5.00	3.00		
3	1.00	4.00	6.00	6.00		
4	1.00	5.00	7.00	4.00		
5	1.00	5.00	9.00	1.00		
6	1.00	6.00	9.00	3.00		
7	2.00	1.00	10.00	2.00		
8	2.00	4.00	8.00	1.00		
9	2.00	5.00	7.00	3.00		
10	2.00	4.00	9.00	2.00		
11	2.00	2.00	10.00	4.00		
12	2.00	5.00	10.00	2.00		
13						
14						

   a. Three Math teachers were assessed by six principals and six peer educators for their teaching effectiveness. Both the principals and peers observed the classroom teaching and completed a Teaching Effectiveness Scale, rating the teachers in terms of planning, instruction, learning environment, and professional practices and responsibilities on a scale of 0 to 10 (0 = very ineffective, 10 = very effective).

b. In this context, the independent variables are (1) the teacher being evaluated (Teacher 1, 2, or 3) and (2) whether the evaluation was made by a principal or peer. The evaluator's rating of teaching effectiveness is repeated measures, dependent variable.

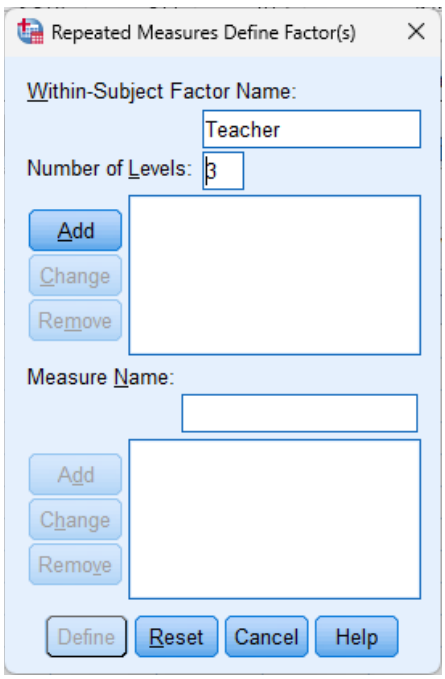
c. The Data View window displays three columns, each corresponding to one of the three teachers: Teacher1, Teacher2, Teacher3. The Rater column indicates whether the evaluator was a principal or peer educator (1 = Principal, 2 = Peer; between-group variable).

CONDUCT RM ANOVA

5. **Analyze** >> **General Linear Model** >> **Repeated Measures** >> This will open the Repeated Measures Define Factor(s) window.

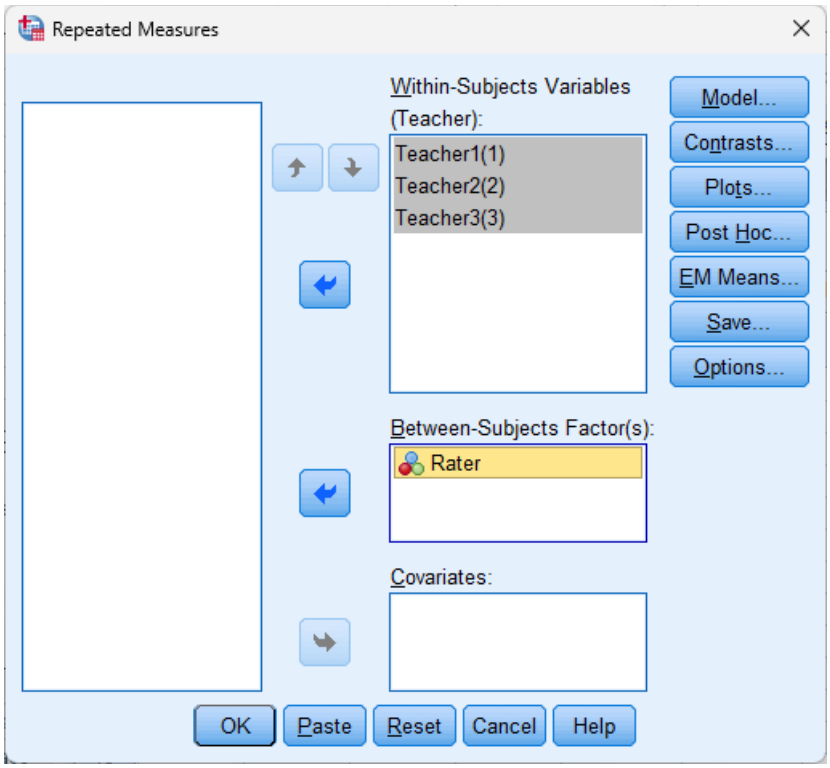
6. Provide a name for the repeated measures (within-subject) variable. Replace "factor1" with "Teacher" in the Within-Subject Factor Name box.

7. Additionally, the number of levels for this variable (i.e., the number of experimental conditions) needs to be specified. Enter "3" into the Number of Levels box >> **Add** >> **Define**.

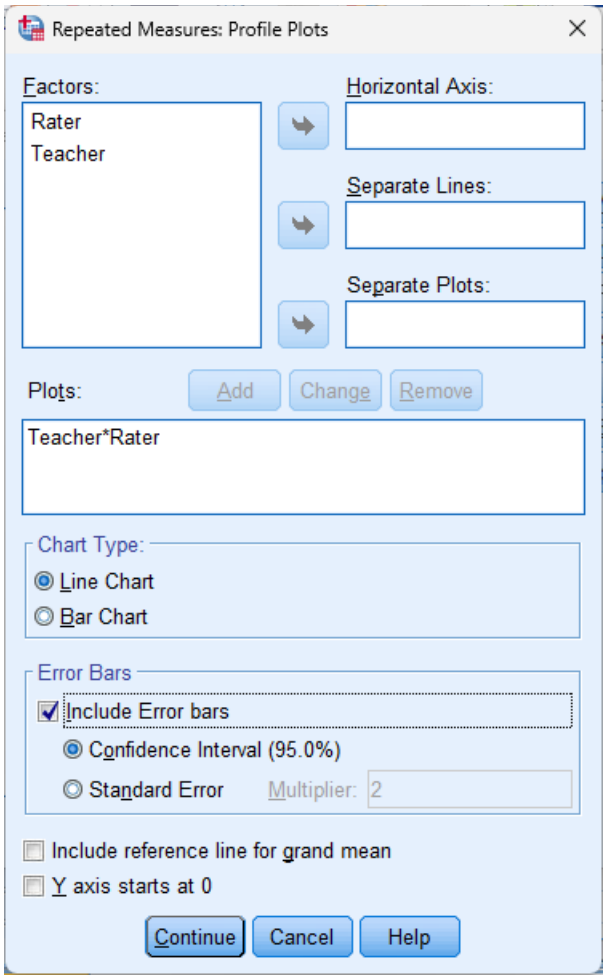


8. The Repeated Measures window has a box labelled Within-Subjects Variables, which displays a list of three question marks each followed by a number. These question marks represent placeholders for the columns that correspond to the three repeated measures. Select Teacher1, Teacher2, and Teacher3 and move these variables into the box.

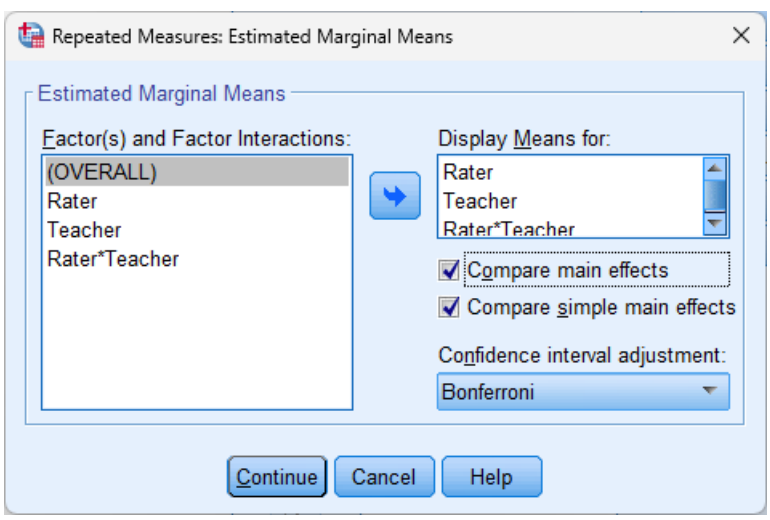
9. Until this point, the procedure is similar to that of repeated measures ANOVA. However, given the mixed design employed for the current study, the analysis must incorporate the *between-group* independent variable. Select Rater and move this variable into the Between-Subjects Factor(s) box.



10. Plotting is a useful tool for visualizing/confirming *interactions* between the variables. **Plots** >> Move the Teacher variable from the Factors box into the Horizonal Axis box >> Similarly, move the Rater variable into the Separate Lines box >> **Add** >> Select **Line Chart** and **Include Error bars** options >> **Continue**.



11. To conduct post-hoc pairwise comparisons of "estimated marginal means", **EM Means** >> Select Rater, Teacher, and Rater*Teacher from the Factor(s) and Factor Interactions box and move this variable into the Display Means for box >> Select **Compare main effects** and **Compare simple main effects** options >> Select **Bonferroni** from the Confidence interval adjustment drop-down menu >> **Continue**.



12. Click **Options** >> Select options including **Homogeneity Tests** >> **Continue**.

13. **Paste** >> **Save** and run the generated syntax.

INTERPRET RESULTS

Mauchly's Test of Sphericity[a]

Measure: MEASURE_1

Within Subjects Effect	Mauchly's W	Approx. Chi-Square	df	Sig.	Epsilon[b] Greenhouse-Geisser	Huynh-Feldt	Lower-bound
Santa	.948	.477	2	.788	.951	1.000	.500

Tests the null hypothesis that the error covariance matrix of the orthonormalized transformed dependent variables is proportional to an identity matrix.

a. Design: Intercept + Rater
Within Subjects Design: Santa

b. May be used to adjust the degrees of freedom for the averaged tests of significance. Corrected tests are displayed in the Tests of Within-Subjects Effects table.

14. SPSS prints out the result of *Mauchly's test*, which assesses whether the data violates the assumption of sphericity. Mauchly's test indicates that the data satisfies the assumption of sphericity, $\chi^2(2) = 0.48$, $p = .79$.

Levene's Test of Equality of Error Variances[a]

		Levene Statistic	df1	df2	Sig.
Teacher1	Based on Mean	.207	1	10	.659
	Based on Median	.225	1	10	.646
	Based on Median and with adjusted df	.225	1	9.939	.646
	Based on trimmed mean	.203	1	10	.662
Teacher2	Based on Mean	2.232	1	10	.166
	Based on Median	1.923	1	10	.196
	Based on Median and with adjusted df	1.923	1	8.887	.199
	Based on trimmed mean	2.223	1	10	.167
Teacher3	Based on Mean	1.025	1	10	.335
	Based on Median	1.250	1	10	.290
	Based on Median and with adjusted df	1.250	1	8.767	.293
	Based on trimmed mean	1.146	1	10	.310

Tests the null hypothesis that the error variance of the dependent variable is equal across groups.

a. Design: Intercept + Rater
Within Subjects Design: Teacher

15. It is also important to check the assumption of homogeneous residual variances using Levene's test. The result of Levene's test suggests that the residual variances are homogenous (i.e., relatively similar for both principal and peer evaluators) across all three repeated measures (Teacher 1, Teacher 2, Teacher 3).

    a. If any of the *p*-values indicates significance (i.e., < .05), it would have compromised the accuracy of the *F*-test for Rater. In such case, consider (i) transforming the data to stabilize the variances between the two groups or (ii) using non-parametric methods.

16. When the model includes an interaction effect, check out the interaction first.

a. If the interaction effect is significant, report and interpret the interaction effect *only*. This is because when there is a significant interaction effect, interpretation of the main effects comprising the interaction could be misleading.

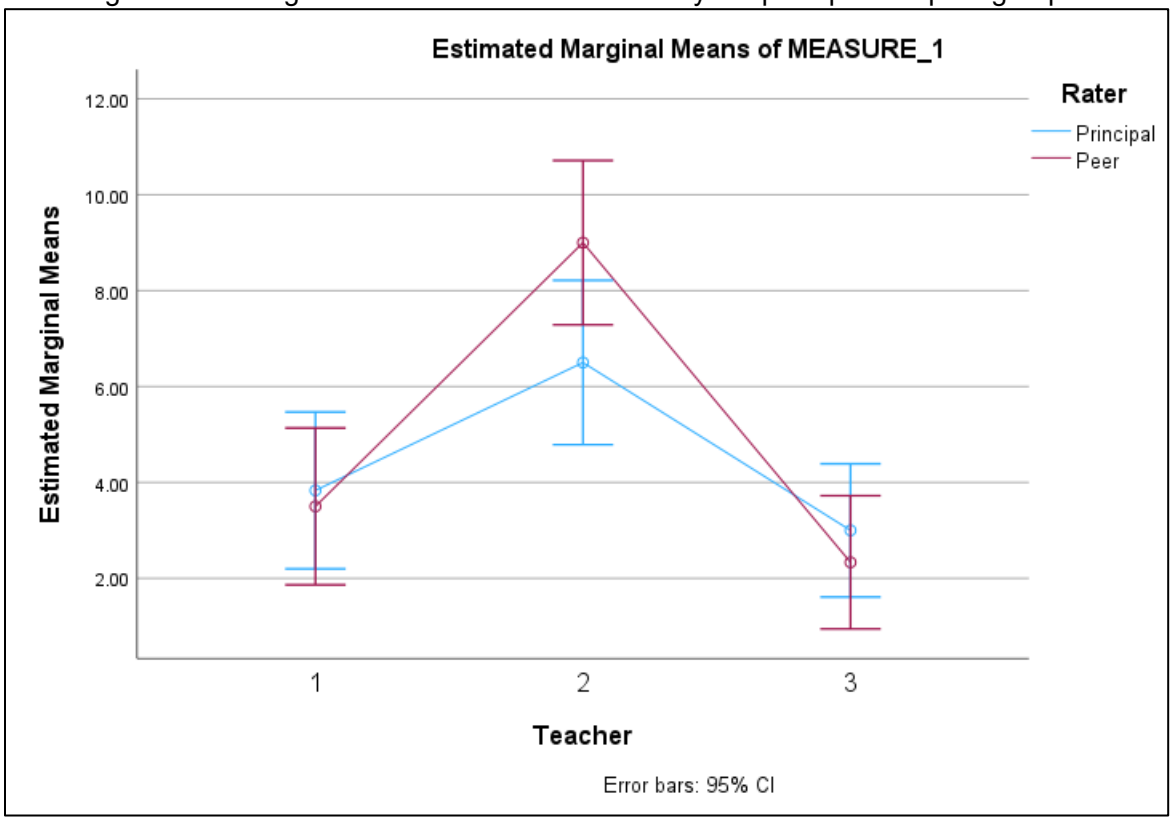
b. If the interaction effect is not significant, report *both* the main and interaction effects.

Tests of Within-Subjects Effects

Measure: MEASURE_1

Source		Type III Sum of Squares	df	Mean Square	F	Sig.	Partial Eta Squared
Teacher	Sphericity Assumed	174.056	2	87.028	36.946	<.001	.787
	Greenhouse-Geisser	174.056	1.902	91.519	36.946	<.001	.787
	Huynh-Feldt	174.056	2.000	87.028	36.946	<.001	.787
	Lower-bound	174.056	1.000	174.056	36.946	<.001	.787
Teacher * Rater	Sphericity Assumed	18.167	2	9.083	3.856	.038	.278
	Greenhouse-Geisser	18.167	1.902	9.552	3.856	.041	.278
	Huynh-Feldt	18.167	2.000	9.083	3.856	.038	.278
	Lower-bound	18.167	1.000	18.167	3.856	.078	.278
Error(Teacher)	Sphericity Assumed	47.111	20	2.356			
	Greenhouse-Geisser	47.111	19.019	2.477			
	Huynh-Feldt	47.111	20.000	2.356			
	Lower-bound	47.111	10.000	4.711			

17. The Teacher-by-Rater interaction is significant, $F(2, 20) = 3.86$, $p < .05$, partial $\eta^2 = .28$, indicating that the ratings on the three teachers varied by the principal and peer groups.



18. The graph of *estimated marginal means* provides insights into the nature of this interaction. The plot shows that the ratings of principals and peers are similar for Teacher 1 and Teacher 3. However, for Teacher 2, peer ratings are noticeably higher than principal ratings. Thus, while

both principals and peers agreed that Teacher 2 performed best, peers gave higher ratings to that teacher compared to principals.

End of Demonstration